History of Changes

Version	Date	Main Changes
1.1	13.02.2017	• Change the following partner's short-name: IPSAS to FU SAV and VTG to VERTECH, in accordance with their official abbreviation on ECAS. Updating of the Gantt chart based on the changes in PartA. • Correction of the total other direct costs for ECODUNA (in section 3.4 Resources to be committed/Table 3.4b). The correct total amount is 100.000 EUR. • Remove of subcontracting cost for SUN and ENEA (in Section 3.4 Resources to be committed and section 4.2 Third parties involved in the project) as explained in Grant Agreement Preparation (GAP) document. • Change the other direct costs for NTUA (in section 3.4 Resources to be committed/Table 3.4b) as explained in Grant Agreement Preparation (GAP) document. • Modification of the linked third parties for PNO Innovation (Section 4.2). • Minor Corrections (typos etc).
1.2	07.11.2017	• Replacement of ECODUNA PRODUKTIONS-GMBH with EPARELLA GMBH • Extension of deadline-date for Deliverables D2.1 (M12 to M15), D2.2 (M12 to M17), and D2.5 (M16 to M18). • Regarding the Installation of the Integrated System inside a suitable "Warehouse" instead of "Green House": change the word "Green House" in the document • Regarding the Cone Construction Material: change the word "steel" to "metal" in the document

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1. Excellence

1.1. Objectives

The VALUEMAG project aims to exploit the convergence between science, chemistry, biology, engineering and biotechnology tools for the creation of new knowledge and innovative applications, with **the main objective to develop an advanced magnetic method for micro-algae cultivation**. Moreover, the implementation of this knowledge in focus on several fields such as the use of micro-algae for **food, cosmetic and nutraceutical products.** A synergistic approach will be used, in order to define a self-sustainable system and have an outcome that is in accordance with the principles of the circular economy.

Targeting the dramatic reduction of cost on micro-algae cultivation we have conceived a technique and a method, which has been recently filed for patent pending, able to cultivate algae in a minimum possible cost. The objectives of VALUEMAG project in this respect are:

- Set up a method to introduce SPANs of 10 nm $\pm$2 nm diameter in micro-algae cell protoplasm of different microalgae species to obtain magnetic modified micro-algae (MAGMA)
- Immobilization of MAGMA using magnetic gradient, due to a soft magnetic cone (SOMAC) of 6 m diameter and ~6 m height; SOMAC is activated with a moderate energy in the order of 10 kW per SOMAC, by using soft magnetic steels after proper welding, roughness forming, annealing and stress annihilation
- Develop economic & viable magnetic Photo-BioReactors (mPBR) for fast growing and easy harvesting of biomass, using the above mentioned SOMAC covered with immobilized MAGMA and thin water layer with minimal extra energy of additional 1 kW per SOMAC, targeting a total of 11 kW of energy requirement per SOMAC

Targeting the production of added-value products from harvested micro-algae, we set further objectives as:

- Implement methods for holistic food production from microalgae biomass
- Extraction of commercially valuable products (nutraceutical, cosmetics) from micro-algae biomass, including a selective magnetic separation method for better, faster and cheaper extraction process
- Develop CO_2 capturing and water re-cycling methodologies based on mPBR

The introduction of **SPANs** into algal cells (known as transformation) will be performed by existing protocols that do not require expensive equipment and chemicals (e.g. electroporation).

The cell-immobilization that will be realized at NTUA in the frame of VALUEMAG project is achieved by introducing SPAN into the microalgal cell protoplasm that will confer them magnetic properties. MAGMA cells then are spread on a magnetized SOMAC surface.

SOMAC will be developed from mild metal using welding methods. The formation of the magnetic surface will be realized using a novel electromagnetic forming method to permit geometrical and stress gradient, permitting magnetic field gradient on its surface, with maximized permeability following stress annihilation methods. The SOMAC will host a pipeline that drives water on top of the cone. The SOMAC slope will be designed in order to avoid turbulence on water flow that may destabilize the MAGMA cell monolayer. Thickness of water will be adjustable and uniform all over the SOMAC surface and in any case it will form a thin film on top of MAGMA. The VALUEMAG consortium aims to solve the microalgal cell cultivation drawbacks thanks to the novel magnetic immobilization of MAGMA.

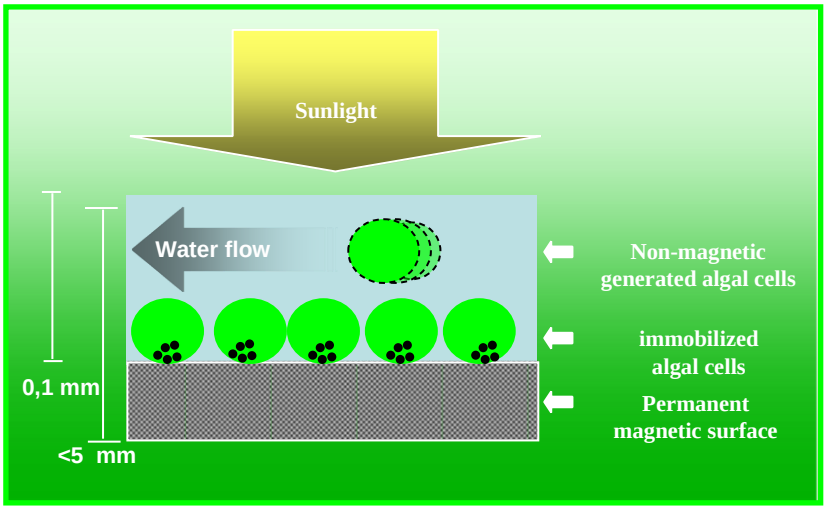

Figure 1.1: Schematic representation mPBR's section, proportions between objects are not real

In fact, the magnetic cell-immobilization of microalgae cells (Figure 1.1) presents many major advantages:

- Reduces the required volume of water, needed for algae cultivation
- Allows for reproducible and standard cultivation conditions (light, temperature, gas exchange, mixing)
- Allows large-scale rapid cultivation independent on the microalgal species
- The parent cell population remains immobilized at each cell division, since SPANs are not removed from them and are polarized facing the magnetic surface, while the new generated ones are right away harvested by simple mechanical methods, such as filtration, a method leaving optimal growth condition
- The shape of the magnetic panel can be adapted easily to any needs since it is light and malleable/ductile
- The mPBR and SOMAC can be productive also in regions with poor water availability that in general present good light exposure (e.g. nearby deserts)
- Both SPANs and SOMAC can be re-utilized after each cycle of biomass production (estimated in ~2 months)
- Contamination with unwanted algae or other microorganisms is unlikely due to: the implementation of a suitable warehouse to include inside the mPBR as well as SOMAC; the fact that each SOMAC represents a single unit not connect with other mPBR, therefore, in case it happens, it can be easily eliminated and
- No oxygen saturation (usually associated with tubular systems) is reached. Indeed, oxygen produced by algae cells can be released easily from each single mPBR unit unlike tubular system where it has to go through dozens meters of pipe system before releasing.

Thanks to these advantages, operating costs for biomass production will be drastically cut down. In particular, limited amount of land/marine/lake area are required because the photo-bioreactor chambers are designed to be developed in vertical contrary to the actual open pond systems which extend horizontally. The amount of water needed to naturally cultivate microalgae is dramatically reduced. Furthermore, after removing biomass, water can be re-introduced in the system because it is clean and no additional treatment must be performed (except when exposed to CO_2). Since volume of the liquid culture is limited, cells concentration per volume unit will be higher with respect to the actual production systems: thus a single step dewatering (e.g. filtration) for biomass collection is required, with evident benefit for the running costs. In addition, no culture contamination with other microorganisms is expected to occur, since the photo-bioreactor chambers are a closed system. Therefore, growth parameters can be easily adjusted and controlled for an optimal cell growth and parental population of microalgae cells (biomass generating) is protected from weathering (wind, rain and dust).

For the extraction of natural products, a new technique based on the use of supercritical CO_2 (supercritical fluid extraction) is preferred since it offers several advantages. In many cases, it allows for lower time extraction, costs and risks compared to traditional techniques with liquid solvents. Many primary/secondary metabolites contained in micro-algae biomass are extremely promising for potential applications in the fields of human and animal nutrition as well as in the production of molecules of pharmaceutical interest. However the cost extraction in many cases is

prohibiting actual production. The VALUEMAG proposed improvement of preparation of primary/secondary metabolites from micro-algae will provide a robust solution to solve this problem. Furthermore, our new selective magnetic separation method will also be used, either in combination with the supercritical CO_2 fluid extraction method or standing alone, in order to provide a new inexpensive and fast method for high-purity natural high added value products from algae cells.

Water recycling, CO_2 capturing and biomass by-product reutilization will strongly depend on different factors: technological, economic, environmental and legislative which will evaluate the feasibility of the technologies; the performance over the different algae system; legislative issues; and technical risks. A combination of different methods and the most innovative solutions will be also studied. The methods that will be analyzed will be, but not limited to: scrubbing of CO2 from the flue gas, mixing the flue gas outlet stream in a bubble column reactor and introducing into micro-algae system, conventional systems (absorption with amines, PSA cycles, and others). Furthermore, in order to maximize the profit of the biomass produced by autotrophic microalgae, the organic residues produced after the extraction of value molecules will be tested and used to grow heterotrophic microalgae that synthesize others high added-value molecules. This will be possible because the extraction of natural products is done using supercritical CO_2 that has no toxicity.

1.2. Relation to the work program

The scope of the Bio-Based Industries Joint Undertaking (BBI JU) is to implement a programme of research and innovation activities in Europe that will assess the availability of renewable biological resources that can be used for the production of bio-based materials, and on that basis support the establishment of sustainable bio-based value chains. Those activities should be carried out through collaboration between stakeholders along the entire bio-based value chains, including primary production and processing industries, consumer brands, SMEs, research and technology centres and universities.

The two main objectives of the BBI JU, as mentioned in article 2 of the Council Regulation are: (a) to contribute to the implementation of Regulation (EU) No 1291/2013 and in particular Part III of Decision 2013/743/EU, (b) to contribute to a more resource-efficient and sustainable low-carbon economy and to increasing economic growth and employment, in particular in rural areas, by developing sustainable and competitive bio-based industries in Europe, based on advanced biorefineries that source their biomass sustainably.

In the Frame of the BBI JU the call BBI-2016-R09 “Exploiting algae and other aquatic biomass for production of molecules for pharma, nutraceuticals, food additives and cosmetic applications” sets several specific scopes that VALUEMAG project directly addresses as following:

Specific scope of BBI-2016-R09	How Project addresses the issue
Proposals should address upstream (adaptation and selection of varieties, growing conditions) and downstream (separation, purification and/or extraction of compounds) processing steps, with the overall aim to maximise the production of compounds of interest.	The VALUEMAG project is focusing on the development of a new concept for magnetically enhanced cultivation of algae, including marine algae, in a far more cheaper way, (from 5-12 E/kg to <0,3 E/kg of algae), so that bringing it very close to the market, with environmentally (water cleaning & CO_2 capturing) and socially (inexpensive production) sustainable issues involved. Salt water is the main medium for algae growth in the execution of this project. However, depending on the microalgae species fresh water is also to be used for algae cultivation. Furthermore, by using Supercritical Fluid extraction and the new magnetic selective separation both quality and quantity of primary and secondary metabolites will be enhanced.

The projects should cover any Technology Readiness Level (TRL) from 3 to 5. In the case of a pilot scale project (TRL 5), proposals should present a credible cost estimate for the proposed processes with a preliminary assessment of their competitiveness when scaled up.	The technology will be based on a soft magnetic metal cone (SOMAC) concept which actually belongs to the marine and maritime technologies. An estimation of the pilot plant based on SOMAC is also provide in this document. The actual TRL of the VALUEMAG is around 3 and targets TRL 5 by the end of the project.We have experimentally and theoretically proved the principle of operation by experiments in lab scale.
Proposals should also include an environmental and socio-economic assessment, for example with an LCA. In particular, when targeting TRL 5, proposals should include an LCA in order to evaluate the environmental and socio-economic performance of the developed processes.	In the VALUEMAG project there is nothing to be wasted: all cultivated micro-algae will be used to develop useful products, the used water will be self-cleaned due to the cultivation procedure, while the space needed is very much limited, as described in the Concept Section. The new concept can be accommodated in several sites, from a maritime environment to a desert. Furthermore, the whole set-up will serve as a research & development laboratory for the improvement of the maritime and algae technology. Project will assess the different impacts on the project, e.g. LCA and will determine the environmental benefits of the VALUEMAG approach. The communication and dissemination plan will aim at spreading the knowledge generated in the project and enhance the training and education in the area. Finally, with the green energy generated from biomass transformation, VALUEMAG bio-refinery will be energetically self-sustained.
Proposals should prove their industrial application taking into account existing EU regulatory framework.	The VALUEMAG project addresses the main issue of the dramatic reduction of the production cost of micro-algal cells and at the same time offers solution for large-scale microalgae cultivation; apart from the cost issue, VALUEMAG targets a variety of micro-algae species with energy and other cost optimized conditions as well as water and CO_2 use; furthermore, all required processes for micro-algae de-watering, nutraceutical and cosmetics manufacturing will take place in a multi-facility bio-refinery

1.3. Concept and approach

The overall approach and methodology of the VALUEMAG project refers to:

- Cost effective cultivation of micro-algae. In fact, the vision of the VALUEMAG project is to decrease the cost of cultivation from 5-12 Euros/kg, as declared by the most recent studies, down to 0.3 Euros/kg, as it is estimated and proven in Chapter 2 (Impact). This is achieved by magnetic techniques in three different levels by using:
    - magnetic nanoparticles to make magnetically responsive microalgae
    - magnetic cultivation cone with large effective surface to trap and cultivate a single layer microalgae cells and minimizing water on magnetic surface cone that results in fast and inexpensive harvesting, as well as
    - implementation of the selective magnetic separation for faster, better (purity) and cheaper extraction of valuable products extraction from micro-algae cells
- The most optimum way of treating algae in order to: ○ produce many different types of expensive nutraceuticals and other valuable products ○ be used for environmental applications (water cleaning and CO_2 capture)

- The most effective ways of developing strategic synergies for algae implementation in engineering by the:
  - Realization of a business plan for the whole activity, including the expected benefits for the main inventors, the current consortium and the European economy
    - Demonstration and road show of the method, the pilot plant and the algae products
    - Selection and agreement with the most appropriate strategic partners, including Classification Societies and representatives from the coastal communities

The technology readiness level (TRL) of all involved technologies and systems are from 3-5, as it will be proven in the next sections, with the final micro-algae cultivation system at the end of the VALUEMAG project, reaching at least TRL 5 (system/subsystem development).

All partners ensure that there will be no sex or gender discrimination in any stage of the project and will respect the indications detailed in http://ec.europa.eu/research/science-society/gendered-innovations/index_en.cfm. Employment of female Chemists, Biologists, Physicists and Engineers from several partners (more than the corresponding male colleagues) has already generated an excellent team work environment, promising innovation and break-through in high end methods and products.

Figure 1.2 illustrates in one scheme the summary of the whole concept-vision of the project.

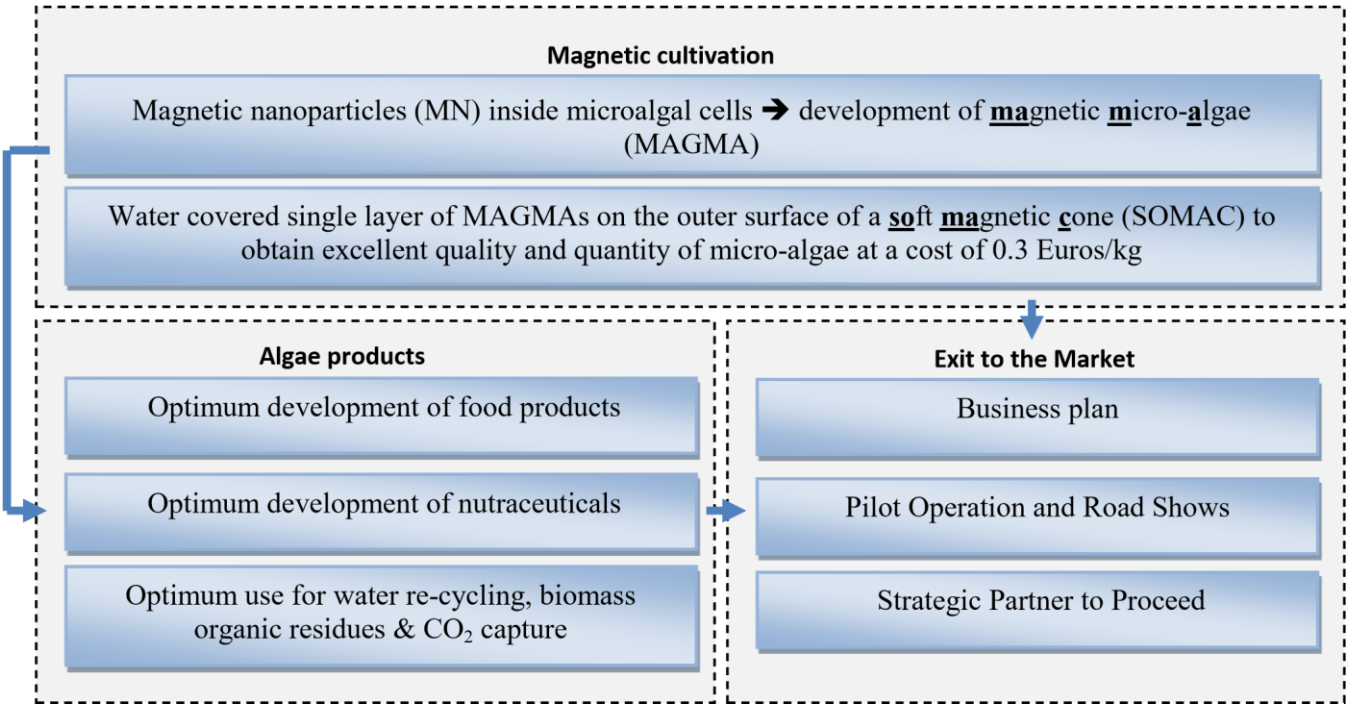

Figure 1.2: The concept – vision of the VALUEMAG project

Introductory

It is estimated that by the 2050 the world population will exceed 9 billion of people from the current 7 billion one[1]. This population growth increases demand, among other human needs, for energy, food, recycled nutrients as well as water. This demand must be sustainable, safe and, most importantly, respecting the planet environment without increasing pollution and/or greenhouse gases such as carbon dioxide (CO_2) and nitrogen oxides (NO_x).

[1] U.S. Census bureau, International Data Base, July 2015 Update. http://www.census.gov/population/international/data/worldpop/graph_population.php

A lot of effort is put world-wide in searching of alternatives to the current sources of energy, food and clean water as well as reduction of pollutions. Though this research represents a tough issue, interdisciplinary research projects like VALUEMAG have the potential to offer a valid alternative. In fact, these problems (demand of clean renewable energy, food, water and greenhouse gases reduction) can have a unique solution: **large-scale rapid biomass production**.

To achieve this goal, among the most promising technologies available, the best opportunity is represented by micro-algae, which can be defined as microscopic solar-powered carbon dioxide-sequestering and nitrogen-fixing organisms. Micro-algae are single-celled life forms; they live in solitary cells (average cell diameter 5-10 μm) or in groups of single cells connected together. Their natural environment is typically freshwater and marine systems.

Most of microalgae species are autotrophic. Under natural growth conditions they primarily require three components to produce biomass: water, CO_2 and sunlight. In fact, via photosynthesis they can generate organic molecules and release oxygen using inorganic elements plus light energy. Micro-algae produce approximately onehalf of the oxygen generated on earth while simultaneously consuming CO_2 during photosynthesis and fixing NOx during their anabolism. Moreover, biomass from micro-algae is a promising source of primary/secondary metabolite products with considerable use in the aquaculture, food additive industry, bio-fertilization, pharmaceutical and cosmetic industry[2]. They can also perform biodegradation of organic pollutants and wastewater treatment. Compared to land crops, microalgae are much faster growing, do not require arable soil, costly irrigation and fertilizers and their growth is not season-dependent under appropriate conditions. Furthermore, biomass generated by microalgae is relatively easy to be collected and transformed since it represents an homogenous raw material that does not need several costly downstream processes due to the absence of macrostructure (e.g.: vascular systems, leaves, roots, bark etc.) as compared to terrestrial crops. Although large-scale microalgae cultivation dates back to 1960, major drawbacks of microalgae production still remain[3].

Among them, costs of cultivation (e.g. liquid culture must be constantly mixed), harvesting and infrastructure are the most important because they represent the bottleneck steps that still obstacle the economically viable of microalgae biomass overall for energy production[4].

Figure 1.3. Raceway Ponds, Eartyrise Spirulla Production plant, California

Attempts to improve microalgae biomass productivity and collection impacted in the past the biology of these organisms, by operating on their genetics, but also biotechnological engendering as well as the design of energy efficient systems to isolate/extract microalgae cells from water (concentration of microalgae cells with the actual technologies is in the order of mg/l). Microalgae cultivation requires specific environmental conditions including temperature, light intensity, mixing and gas exchange. Different species have slightly different requirements. Table 2 shows a comparison of the existing commercial microalgae cultivation technologies with respect to the system proposed by VALUEMAG project. Currently, over 90% of world microalgae biomass production is realized in large raceway ponds (also known as racetrack) (Figure 1.3)[5][6].

It is clear that the magnetic methodology proposed within the VALUEMAG project, becomes effectively the solution for the sustainable development of large-scale micro-algae cultivation.

[2] Rosenberg, J. N., Oyler, G. A., Wilkinson, L., & Betenbaugh, M. J. (2008). A green light for engineered algae: redirecting metabolism to fuel a biotechnology revolution. Current opinion in Biotechnology, 19(5), 430-436

[3] Mata, T. M., Martins, A. A., & Caetano, N. S. (2010). Microalgae for biodiesel production and other applications: a review. Renewable and sustainable energy reviews, 14(1), 217-232

[4] Kim, J., Yoo, G., Lee, H., Lim, J., Kim, K., Kim, C. W. & Yang, J. W. (2013). Methods of downstream processing for the production of biodiesel from microalgae. Biotechnology advances, 31(6), 862-876

[5] Safarik I, Maderova Z, Pospiskova K, et al. (2014). Magnetic decoration and labeling of prokaryotic and eukaryotic cells. In: Fakhrullin RF, Choi I, Lvov YM, eds. Cell surface engineering: fabrication of functional nanoshells. RSC Smart Materials No.

[6] Cambridge: RSC, 185–215

	Capital Cost	Running cost	Productivity	Reliability
Ponds	●	●	●	●●
Raceway Ponds	●●●	●●	●●●	●●●
Cascade System	●●●	●●	●●●●	●●●
Tubular PBR	●●●●●	●●●●	●●●●	●●●
Fermenter	●●●●●	●●●●●	●●●●●	●●●●●
VALUEMAG (estimation)	●●●	●	●●●●●	●●●●●

Table 1.1: Comparison of the existing commercial microalgae cultivation technologies vs VALUEMAG

Production of energy (biodiesel, syngas), for example, from microalgae with this method is discouraged by initial capital investment and running cost which are the most critical drawbacks for an economically profitable production of biomass. The major criticism concern, as above mentioned, the costs for cultivation/harvesting and the cost of infrastructure. In particular, open ponds, raceway and cascade systems require large amount of land and water to cultivate microalgae.

The major disadvantages associated with these open systems, partially determining the low productivity, are:

- Land costs when commercial use microalgae, e.g. natural products extraction, is intended
- Contamination by other algae, bacteria and/or other microorganisms
- Water evaporation as well as large volumes of water
- Difficult to maintain an open algal culture during the rainy season
- Energy waste for constant culture mixing
- Very low micro-algae cell concentration in the liquid culture.

Figure 1.4. Chlorella Photobioreactors system. Roquette GmbH- Klötze, Germany

The last drawback adds extra expenditure because large amount of water must be treated in order to concentrate microalgae biomass. An optimal harvesting technique should be independent of the cultured species, consume little energy and few chemicals and not damage the valuable products in the extraction process. Traditional harvesting method may involve up to two steps:

I) bulk harvesting (known as primary harvesting)–to separate microalgae from suspension, such as sedimentation, flocculation, and flotation; and

II) thickening (known as secondary dewatering)– to concentrate the micro-algae slurry after bulk harvesting, such as centrifugation and filtration[7].

The use of closed systems solves many of the above mentioned problematic but yet the infrastructure remains expensive as does the running cost. Tubular PBRs (Figure 1.4), for example, possess two regions, light-illuminated surface and dark region inside of reactor. For efficient light transfer to microalgal cells, mixing of large volumes of water is required to circulate cells between dark and bright region of PBR in addition to optimization of light path length and shape of PBR. Whereas, concerning fermenters

[7] Zhang, X., Rong, J., Chen, H., He, C., & Wang, Q. (2014). Current status and outlook in the application of microalgae in biodiesel production and environmental protection. Frontiers in Energy Research, 2, 32

they perform good biomass production but utilizing microalgae that are heterotrophically cultivated micro-algae, they require appropriate organic carbon sources that involve extra-cost.

Magnetic Cultivation Efficiency

With the advent of nanotechnologies new tools became available providing additional functionalities. In particular the biotechnological use of magnetic nanoparticles is currently investigated mainly for microalgae cells harvesting[8]. Notably, magnetic modifications of microalgae cells concerned, until now, the attachment of magnetic nanoparticles only onto the cell surface[5].

In order to really improve the microalgae biomass productivity and lower its cost production, VALUEMAG propjet proposes a new kind of microalgae magnetic modification by **introducing inside the micro-algae cell-body** SPANs that will allow to **immobilize them in a thin monolayer on top of a magnetic surface with the shape of ultra-thin flat panel** (mPBR).

Once MAGMA are attached onto a magnetic surface, a thin flow of water is appropriately circulated over them in order to mix the liquid culture and at the same time harvest the generated cells (biomass).

- **Introducing magnetic nanoparticles into microalgae**

The specific challenge at the VALUEMAG project is the magnetic immobilization of different micro-algae species for large-scale biomass production in technological advanced bio-refineries. The concept of algal cells immobilization for industrial purposes has been investigated in the past, and abundant literature exists[9], but chemicals and reagents used for cell-immobilization yet represent a barrier for scaling up facility based on these biotechnologies. So far, immobilized microalgae cells have been used mainly in small devices to reduce organic and inorganic pollutions as well as biosensors. Even tough for the vast majority of microalgae species studied the cell-immobilization does not affect the growth rate (similar to free cells and in the same case even better), no large scale facilities for biomass production utilize immobilized microalgae cells[8].

SPANs are synthetic metal-based nano-material (Fe or Fe_3O_4 cores covered by organic thin layers) with magnetic properties. Our technology to produce such SPANs is either the chemical vapour deposition (CVD) or the flame spray pyrolysis technique, according to which the size and stoichiometry of the produced particles can be well controlled. The size of SPANs produced with CVD can be in the order of 5-100 nm in diameter, with size uncertainty of $\pm$1 nm, with productivity of some gr per hour, while flame spray pyrolysis offers larger quantity (kg per hour) in the cost of a larger uncertainty of dimensions. Our (NTUA) brand-new microwave assisted flame spray pyrolysis is promising acceptable uncertainty ($\pm$2nm) and productivity of kg per hour. SPANs synthesis, storage, and handling costs are relatively low and high quantity can be easily and rapidly produced. Moreover, superparamagnetism, a feature that is important for SPAN usability consisting in the fact that after eliminating the magnetic field the particles no longer show magnetic interaction, offers the key to safely cultivate micro-algae. Indeed, the presence of SPANs in microalgae cells will not interfere with the biology of these organisms since only when required SPANs are magnetized. During all MAGMA preparation no magnets are used and after SPANs introduction, micro-algae can be cultivated in standard condition (without magnetic field) in order to allow for efficient recovering. Methods that allow the introduction of SPANs inside microalgae cells may stress them and a short-time period, 1-5 hours, is required before to get a proper growth rate. In fact, during the recovering microalgae cells will not be in contact with magnetic surface because cultivated in standard condition.

The introduction of SPANs in microalgae protoplasm (named transformation) will be performed using the most popular methods: electroporation and/or vortexing with glass beads. These methods are widely used for genetransfer (DNA) applications because it enhances the cell-membrane permeability of a variety of microorganisms such as bacteria, yeast, algae, plant and human cells. However, other exogenous molecule like, RNA, proteins and small

[8] Safarik, I., Prochazkova, G., Pospiskova, K., & Branyik, T. (2015). Magnetically modified microalgae and their applications. Critical reviews in biotechnology, 1-11

[9] Moreno-Garrido, I. (2008). Microalgae immobilization: current techniques and uses. Bioresource technology, 99(10), 39493964.

inorganic nanoparticles can be introduced in the cell-body with this technique. Both of them are inexpensive and do not require sophisticated machineries.

The principle electroporation is based on the fact that in presence of a transient electric field cell membranes generate pores that allow for the transfer of molecules inside the protoplasm which are otherwise excluded. Electroporation capitalizes on the relatively weak nature of the phospholipid bilayer's hydrophobic/hydrophilic interactions and its ability to spontaneously reassemble after disturbance. Thus, a quick voltage shock may disrupt areas of the membrane temporarily, allowing polar molecules to pass, but then the membrane may reseal quickly and leave the cell intact. Electroporation is a physical method with excellent reproducibility, low cost and it can be scaled up for high throughput applications. Furthermore, it offers superior transformation efficiency with respect to other chemical and physical methods.

The glass beads method to delivery organic molecules (DNA) into algae cells has been successfully used[10]. This method is usually used as an alternative to the enzymatic digestion of hard cell walls that may surround the cell membrane of microorganisms like bacteria, yeast, algae and plant. The principle is based on soft mechanical damaging of hard cell walls and membrane with glass beads. Damaged cell walls (pores) may allow to pass organic and inorganic molecules that other ways are excluded.

In the best of our knowledge, nobody has used electroporation or glass beads to introduce SPANs in microalgae cells. Nevertheless, abundant scientific literature have been found concerning gene-transfer by electroporation or glass beads and microalgae[11,12,13]. As mentioned before, the nature of cell-membrane pores generated with these methods are not molecule-specific because it depends only on the transient perturbation of the cell-membrane. Indeed, the phospholipid bilayer of the plasma membrane has a hydrophobic exterior and a hydrophobic interior, any polar molecules, including DNA, protein and nanoparticles, are unable to freely pass through the membrane. Therefore once pores are generated, they will allow any kind of molecule including SPANs to pass through cellmembrane. However, modification of SPANs surface, named functionalization, will be explored in order to find the optimal combination between introduction of SPANs and viability of micro-algae cells. Surface functionalization of SPANs determines their interaction with the environment. These interactions ultimately affect the delivery of nanoparticles into the cells by appropriate functional molecules on the particle surface. Furthermore, surface functionalization will prevent the aggregation of the nanoparticles both in the medium used for transformation and inside the microalgae cells. A standard surface modification that results in nanoparticles with increased stability into biological environment consists in the use of poly(ethylene) glycol (PEG). PEG possess amphiphilic properties, and nanoparticles with those molecules can be soluble and stable in a number of solvents, with intermediate polarity such as the protoplasm of microalgae cells, for which we have developed a quality control method to control the percentage of SPAN coverage, a vital property for avoiding toxicity.

Since the composition of microalgae cell walls may vary between different microalgae species, transformation parameters will be optimized for each species. Algal cell walls, as mentioned, are known to consist of multiple layers. Some cell walls are rich in neutral sugars while some contain only trace amounts of sugar. In view of the fact that microalgae cell walls are, similar to a shield, located in the external part of the cell membrane they may interfere with the SPANs transferring process. Nevertheless, recent electroporation application yields a remarkably high transformation efficiency, even when the cell wall is not removed[10]. However, in the frame of research plans of VALUEMAG project transformation experiments will be performed with various concentration of SPANs and both with electroporation or glass beads methods for each species. Viability, toxicity and magnetic properties of micro-algae will be tested in laboratory and once the best set up is found the transformation will performed in a dedicated small facility inside the mPBR.

[10] Feng, S., Xue, L., Liu, H., & Lu, P. (2009). Improvement of efficiency of genetic transformation for Dunaliella salina by glass beads method. Molecular biology reports, 36, 1433-1439.

[11] Qu B. et al (2012). Droplet electroporation in microfluidics for efficient cell transformation with or without cell wall removal. Lab on a chip, 12, 4483-4488

[12] Potvin, G., and Zhang, Z. (2010). Strategies for high-level recombinant protein expression in transgenic microalgae: a review. Biotechnology Advances, 28, 910-918

[13] Shimogawara, K., Fujiwara, S., Grossman, A., & Usuda, H. (1998). High-efficiency transformation of Chlamydomonas reinhardtii by electroporation. Genetics, 148, 1821-1828.

We assume that the parental population of MAGMA will reach a point where no more cell division will be possible. This is due to several reasons. Among them the fact that after a certain number of cell-division microalgae accumulate irreversible damage in their genome, as well as occasional cell-death, due to cell-body damage, are the most important ones. To ensure optimal biomass productivity, not dividing MAGMA will be stripped from the magnetic surface and replaced with fresh-generated MAGMA. Thanks to superparamagnetic properties, SPANs will be recovered from old MAGMA. This will be done by disrupting MAMGA in presence of a strong magnetic field able to trap SPANs. Then, inorganic SPANs will be separated from the rest of organic biomass and used for the most appropriate transformation.

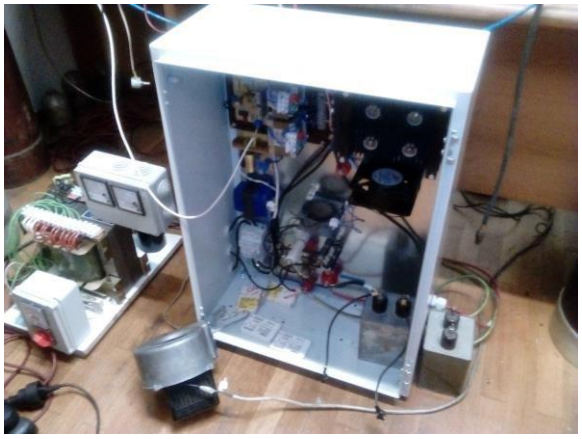

Figure 1.5. Power supply for the electroporation system. The design of this power supply permits a wide range of excitation frequencies from μHz to several tens of □Hz in order to achieve proper electroporation

Magnetic cultivation

The magnetic cultivation method and system is currently pending for patent. A basin fixed inside a suitable warehouse, hosts magnetic micro-algae in a water environment, such environment being either sea or potable water. Transformation allows to introduce SPANs in micro-algae, thus making them MAGMA. After transformation a mechanical pump transmits water enriched by MAGMAs to the peak of SOMAC reactor, with micro or milli curvature roughness realized by the new pending patent non-contact electromagnetic forming method. The basis of the SOMAC is inside the above mentioned basin.

The speed of the MAGMA enriched water is slow enough to avoid turbulence in their propagation. Thus, a thin layer of water with laminar flow is obtained on the SOMAC surface. Magnetic polarization of different areas of the SOMAC results in trapping MAGMAs in these magnetically polarized areas. Apparently, the micro-algae without SPANs drop back in the above mentioned basin, since they cannot be trapped in the SOMAC. Thus, following proper algorithms of magnetic polarization, results in trapping MAGMAs in proper areas of the SOMAC.

The SOMAC and its corresponding basin are fixed inside a greenhouse in order to avoid contamination and also to maintain controlled growth conditions such as temperature, humidity and gas exchange (CO_2 and Oxygen). Furthermore, the thin layer of the MAGMA water coverage is recycled from the basin.

Having MAGMAs under a micro-thin layer of water permits their holistic cultivation: groups of MAGMAs are prohibited to grow, because none of them is hidden by other MAGMAs from sunlight and flowing water. This way, all MAGMAs are multiplied in time in a free space, under optimum conditions. The generated micro-algae without SPANs inside cannot be trapped in the SOMAC and therefore they are gently forced to drop in the mentioned basin and will form the biomass to be collected. At regular interval new micro-algae containing SPANs are to be injected in the system, so that the new MAGMAs are trapped in the SOMAC whereas death or poorly viable MAGMA are eliminated by switching-off the power supply of the SOMAC. The new-generated MAGMAs will have by design zero or negligible amount of SPANs inside them. Thus, there is no doubt that the new MAGMAs will have magnetic moment to permit trapping on the SOMAC surface and therefore they are driven to the mentioned water basin.

These new MAGMAs with zero or poor magnetic moment, can be re-transformed again, in order to re-enter and being trapped by the magnetizing gradient on the SOMAC surface. It is noted that the sea or the potable water used to grow MAGMAs is cleaned under the influence of the MAGMAs, thus suggesting the possibility of a reutilization of such kind of water and –importantly- its introduction in the environment is completely risk-free.

Monitoring the quality of the MAGMAs in this basin, can offer the decision to temporarily stop the MAGMA multiplication and gain back the produced MAGMAs, by centrifugation or/and other techniques. This is the actual way to maximize the micro-algae production.

The most important factor in micro-algae biomass production is light intensity. Micro-algae can only utilize the energy available in the 400–700 nm wavelength range. Light sources can be divided into natural sunlight and artificial cold light that is mainly applied in closed cultivation. The rate of photosynthesis is proportional to light intensity. When irradiance is increased, microalgal growth rate accelerates, but exposure of cells to long periods of high light intensity causes photoinhibition. Besides light intensity, it was found that light–dark cycles could also significantly influence microalgal growth by avoiding sustained exposure to high photon flux density and providing dark time for microalgae to repair photo-induced damage. The light–dark periods for most microalgal cultivation are 24h:0h, 16h:8h and 12h:12h. This varies with microalgal species. The SOMAC surface will be illuminated with natural light and the intensity will be controlled by the design of magnetic cones and Cultivation chamber infrastructure.

As previously mentioned, the productivity of immobilized MAGMA is limited and after a certain period the MAGMA monolayer needs to be replaced. The magnetic surface are then gently scraped and MAGMA cells, still alive though poorly dividing, will be collected. In order to maximize the biomass yield, re-use purposes of SPANs and also for efficient properties of micro-algae (natural ingredients etc.), separation of magnetic nanoparticles from algae is necessary. The removal of magnetic nanoparticles from algae will be realized by a mechanical disruption of micro-algae cells that will be released in the medium all cell component including SPANs. A magnetic field gradient caused by the power supply will be responsible for the magnetic separation of nanoparticles from the rest of microalgae content. Since the quantity of MAGMA cells that need this kind of treatment is not high and the process does not need to be repeated daily, its cost will not influence the management of the photobioreactor.

The main microalgae that will be used in the frame of VALUEMAG project are:

- ***Haematococcus pluvialis*** (Chlorophyceae, Volvocales): is unicellular fresh water microalga distributed in many habitats worldwide. It is considered as the best natural source of astaxanthin and the main producing organism of this commercial product Astaxanthin (3,3′-dihydroxy-ß-carotene-4,4′-dione) is a bright red secondary carotenoid from the same family as lycopene, lutein, and *ß-carotene*,
- ***Chlorella***: it has an oil content in the range of 25-32 wt.% but in some cases could achieve 85 wt.% as for Chlorella pyrenoidosa and C. protothecoides. Lipid types in these microalgae species are mainly triglycerides and then more suitable for biochemicals conversion but also *for dietary supplement;*
- ***Nannochloropsis****:* it is an eustigmatophyte with a triglycerides contents in the range 31-68 wt.% depending of nitrogen content in the culture broth usable for *acquaculture*;
- ***Dunaliella****:* similar to Botryococcus braunii it is a green microalgae that is able to growth in an high range of salts content. It has till to 23% as lipids content and is used in large scale plant for carotenoid productions as *ß-carotene*;
- ***Crypthecodinium cohnii****:* is a species of dinoflagellate microalga that can be used in VALUEMAG for its high omega-3 fatty acid content.
- ***Scenedesmus almeriensis****:* green algae with an high *lutein* content. Lutein is a xantophile recommended for the prevention of some types of cancer, cardiovascular diseases and retinal degeneration. Lutein is also a food dye allowed by EU and reported as E 161 b.

However other kind of species able to synthesize high-value products will be considered.

- **Cosmetics & food (nutraceuticals) – Concept & approach**

Micro-algae are one of the oldest living organism species. Their photosynthesis is similar to that of higher plants, but they are generally more efficient in terms of solar energy conversion. Growing in aqueous media, they use ably water, CO_2 and nutrients. Micro-algae can be photoautotroph, heterotrophic, or mixotrophic. Autotrophic microalgae use sunlight to convert H_2O and CO_2 in O_2 and sugars. They are able to fix CO_2 from different sources, including atmosphere, water-soluble carbonates and industrial exhaust gases. About 1.8 kg of CO_2 are fixed per kg (dry weight) of produced micro-algal biomass.

Micro-algae have also been proposed as feedstock for third-generation biofuel production[14, 15]. Indeed, they have several advantages as compared to terrestrial plants: 1) they grow fast and, potentially, throughout the year, although climatic conditions should be taken into account; 2) they can grow on marginal or desert soils, therefore there is no land competition, in opposition to lignocellulosic biomass; 3) many species can be grown in salt or brackish water, hence the use of fresh water is minimised, even not necessary; 4) their water requirement per kg of biomass produced is lower than that for terrestrial crops; 5) they can be used for CO_2 sequestration from the flue gas from industrial installations; 6) some species accumulate large amounts of triglycerides (30-40% of the dry weight), suitable for the biodiesel production, while others accumulate starch, suitable for the bioethanol production; 7) application of herbicides or pesticides is not required; 8) some species can be used for bioremediation purposes.

Despite these advantages and the presence of many start-up companies in the field, a commercial supply chain for this production has not been developed yet. The main reason is that the cost is still too high for the production of algal biomass. The current commercial applications of micro-algae are mainly in the fields of food supplements, feed industry and cosmetics (Figure 1.6), as proposed in the VALUEMAG project.

Figure 1.6: Commercially cultivated microalgae and their products

Despite the wide use of micro-algae as "healthy food", their positive effects are somewhat controversial: for example, the use of Aphanizomenon as food may involve health risks, due to the toxins present in some preparations[16]. On the other hand, **other species of micro-algae do not imply any risks and are extremely promising for potential applications in the fields of human and animal nutrition, and in the production of molecules of pharmaceutical interest**, as reported below (Table 1.2):

[14] Brennan L, Owende P (2010) Biofuels from microalgae-a review of technologies for production, processing, and extractions of biofuels and co-products. *Renewable and Sustainable Energy Reviews*14: 557-577;

[15] Mata TM, Martins AA, Caetano NS (2010) Microalgae for biodiesel production and other applications: A review. Renewable and Sustainable Energy Reviews 14: 217-232;

[16] Dittmann E, Wiegand C (2006) Cyanobacterial toxins-occurrence, biosynthesis and impact on human affairs. Mol Nutr Food Res 50: 7-17;

Microalgae	Annual production	Producer country	Application and product	Price (€)
Spirulina	3000 tonnes dry weight	China, India, USA, Myanmar, Japan	Human nutrition Animal nutrition Cosmetics Phycobiliproteins	36 kg^{-1} 11 mg^{-1}
Chlorella	2000 tonnes dry weight	Taiwan, Germany, Japan	Human nutrition Cosmetics Aquaculture	36 kg^{-1} 50 l^{-1}
Dunaliella salina	1200 tonnes dry weight	Australia, Israel, USA, Japan	Human nutrition Cosmetics B-carotene	 215–2150 kg^{-1}
Aphanizomenon flos-aquae	500 tonnes dry weight	USA	Human nutrition	
Haematococcus pluvialis	300 tonnes dry weight	USA, India, Israel	Aquaculture Astaxanthin	50 l^{-1} 7150 kg^{-1}
Crypthecodinium cohnii	240 tonnes DHA oil	USA	DHA oil	43 g^{-1}
Shizochytrium	10 tonnes DHA oil	USA	DHA oil	43 g^{-1}

13

Table 1.2: Commercial production of micro-algae

Many micro-algae accumulate high levels of polyunsaturated fatty acids (PUFA) **ω-3**, as eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA). The PUFAs **ω-3** and ω-6 fatty acids are essential in the diet of many animals, including humans, which are unable to synthesize them. In Figure 1.7 is shown some PUFAs **ω-3** and ω-6 in our diet and their food sources:

Figure 1.7: Polyunsaturated fatty acids ω-3 and ω-6 and their food sources. The position of the double bond that distinguishes the two classes of molecules is highlighted by a red circle

Several lines of evidence suggest that humans have evolved based on a diet that had a 1:1 ratio of PUFA ω-6 and ω-3. In the Western diet this ratio has increased to 15:1[17]. This results in a greater activity of the pro-inflammatory responses mediated by arachidonic acid (an ω-6 long-chain), which in turn would explain the increased incidence of cardiovascular diseases and certain types of cancer. A 4:1 PUFA ω-6 and ω-3 in the diet reduces mortality by 70% in the secondary prevention of cardiovascular diseases, while a 2.5:1 ratio reduces the rectal cell proliferation in patients with colorectal cancer or suppresses inflammation in patients with rheumatoid arthritis. In addition, the ω-3 have an important role in brain function: DHA represents 40% of the phospholipids of the membrane in the brain, and studies with nonhuman primates and human infants indicate that DHA is essential for the functional development of the brain, in particular in premature infants[18]. Most of the ω-3 in our diet is present in the form of α-linolenic acid (ALA), a common PUFA in nuts and oil flax (Figure 6) . In the human body, ALA is inefficiently (<10%) converted to DHA and EPA. The main dietary sources of DHA and EPA are fish (tuna, salmon, sardines, herring) and fish oil.

[17] Simopoulos AP (2008) The omega-6/omega-3 fatty acid ratio, genetic variation, and cardiovascular disease. *Asia Pac J Clin Nutr 17 Suppl* 1: 131-134;

[18] Simopoulos AP (2011) Evolutionary aspects of diet: the omega-6/omega-3 ratio and the brain. *Mol Neurobiol* 44: 203-215;

FAO projections (2008) show a stabilization of the world's fish production, with the capture progressively replaced by aquaculture, while world production of fish oil is currently about 1 Mton / year and has a slightly decreasing trend. Fish production by aquaculture presents a sustainability problem: the feed of the most prized fish, such as bream, sea bass and salmon, contains protein and oil derived from caught fish. In general in the aquaculture sector the ratio of caught fish used for feed and fish produced is 0.63, but it goes up to 5.0 for salmon[19]. The growing demand for proteins and fish oil for aquaculture has led to a 30-40% increase of their price in 2012, but the strongest demand for **ω-3** long-chain comes from human nutrition. Assuming a recommended daily dose of 2 grams/day of oil rich in **ω-3** and a world population of 9 billion in 2050, the potential demand for human consumption is 6.5 Mln tons, or 260 billion of the € current cost of 40 €/kg. Of this, only 1 Mtons is likely to be covered from fish oil. It seems evident that alternative sources of low cost will go quickly retrieved.

The **carotenoid** market is estimated at around € 1.2 billion in 2010, with the opportunity to grow to € 1.4 billion in 2018. Among the best-selling compounds are beta-carotene and lutein (used as dietary supplements for the prevention of avitaminosis and the related macular degeneration, a degenerative disease of the retina) and **astaxanthin**, which is used as a food additive in the fishing industry, but also with potential applications as a powerful antioxidant in the human diet[20] (Figure 1.8). Prices vary greatly, from € 20/kg for **beta-carotene**, to € 4.000/kg **astaxanthin.** The current market is mainly dominated by the chemical synthesis of compounds, with the exception of lutein, which is extracted from the petals of marigold and annatto, which is extracted from the seeds of Bixa orellana. Micro-algae are able to synthesize most of these compounds, often in a highly pure form, such as beta-carotene from Dunaliella salina and the produced **astaxanthin** from Haematococcus pluvialis. The goal of VALUEMAG project is to reduce the costs of micro-algae production up to competitive levels with those of synthetic compounds.

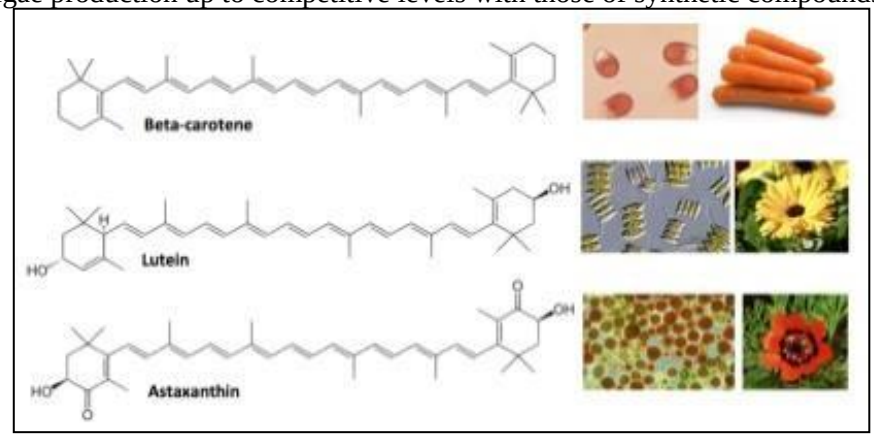

Figure 1.8: Chemistry and natural sources of beta-carotene: carotenoids, lutein and astaxanthin[19]

A potential application of spirulina and micro-algae is the replacement of animal protein as feed, especially those used in aquaculture. The micro-algal proteins have proved of good substitutes for those of animal origin in the diet of farmed fish[21].

Last but not least, the micro-algae are emerging as an alternative system for the drug production. This relatively new platform offers several advantages, including: 1) a short time from the generation of transgenic microalgae to the "scaling-up" 2) rapid growth (doubling time of a few hours) and ease of cultivation, 3) safety, since the microalgae are no vehicle of human pathogens or animal and are generally considered safe, 4) growing in sterile conditions, promoting the production of proteins for human use. The model system Chlamydomonas reinhardtii, whose nuclear and plastid genomes are completely sequenced and easy transformation, has so far been widely used for the

[19] Naylor RL, Hardy RW, Bureau DP, Chiu A, Elliott M, Farrell AP, Forster I, Gatlin DM, Goldburg RJ, Hua K, Nichols PD (2009) Feeding aquaculture in an era of finite resources. Proc Natl Acad Sci U S A 106: 15103-15110;

[20] Greaves R, Petkovich MCON, Miyazawa TCON, Albertini B (2012) *Vitamin A and Carotenoids: Chemistry, Analysis, Function and Effects*. Royal Society of Chemistry;

[21] Olvera-Novoa MA, Dominguez-Cen LJ, Olivera-Castillo L, Martinez-Palacios CA (2008) Effect of the use of the microalga Spirulina maxima as fish meal replacement in diets for tilapia, Oreochromis mossambicus (Peters), *fry. Aquaculture Research* 29: 709-715;

production of proteins of pharmaceutical interest[22]. Other microalgae, with a potential use in the animal feed, are receiving attention as vehicles for the production of oral vaccines against the most common animal diseases.

The main obstacle to micro-algae use is the current production cost, calculated at around € 5 kg of dry biomass[23]. This cost have to be reduced, at least 10 times, while it is already compatible with the production of other products, such as food additives, molecules for the fine chemicals, proteins of pharmaceutical interest, animal vaccines at low cost. Research in these areas is fraught with interesting results in the short and medium term.

For many years, synthetic fungicides have been used for control plant pathogenic fungi. However, the extensive use of these chemicals led to the development of resistance in many areas around the world[24]. In order to overcome this problem, higher concentrations of these chemicals were used, but these concentrations increase the risk of highlevel toxic residues in the products. Thus, there has been a growing interest on the research of the possible use of plant secondary metabolites for pest and disease control in agriculture[25][26]. Biological control has proven to be effective against a variety of pathogenic fungi and bacteria[27,28]. Micro-algae are rich sources of structurally new and biologically active metabolites. In recent years, there have been many reports of algae derived compounds that have a broad range of biological activities and are explored for use in agriculture, such as biocontrol agents against plants diseases. In these studies it was shown that the extracts of algae have a dual effect: they strongly reduce the growth of fungal mycelium and stimulate biochemical resistance in plants[29,30]. **Environmental aspects (water recycling and CO_2 capture)**

Eutrophication of water systems (abnormal accumulation of aquatic flora that obstacle/compete with other form of aquatic life) such as rivers, lakes, seas and oceans is an inevitable consequence of all contemporary societies. The main cause of eutrophication are domestic and agro-industrial wastewaters that release large amounts organic phosphorus and nitrogen that feed aquatic flora. Moreover, most wastewaters are usually hazardous to human populations and the environment (e.g. heavy metals) and must be treated prior their disposal in water bodies. A good practice is to remove most nutrients and pollutions prior to discharge wastewater into environment, but, though usually obligatory, is not performed because of the high costs. In addition, removal of nutrients such as P or N along with heavy metals is usually carried out in tertiary treatment through advanced methods. The use of these technologies (chemical precipitation, ozonation, reverse osmosis and carbon adsorption, amongst other) implies higher costs in the sewage treatment, in the order of 4-16 times more expensive in comparison with the primary treatment.

Bio-treatment with microalgae is particularly attractive because of their photosynthetic capabilities, converting solar energy into useful biomasses, which could be uses for production of fine chemicals, and incorporating nutrients such as nitrogen and phosphorus causing eutrophication. The algal systems can treat human sewage, livestock wastes, agro-industrial wastes and industrial wastes[31]. A complete tertiary process aimed at removing ammonia, nitrate and phosphate will thus be about four times more expensive than primary treatment. Microalgal cultures offer an elegant solution to tertiary and quinary treatments due to the ability of microalgae to use inorganic nitrogen and phosphorus

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[22] Specht E, Miyake-Stoner S, Mayfield S (2010) Micro-algae come of age as a platform for recombinant protein production. *Biotechnol Lett.* 32(10):1373-83;

[23] Norsker NH, Barbosa MJ, Vermue MH, Wijffels RH (2011) Microalgal production-a close look at the economics. *Biotechnol Adv* 29: 24-27.

[24] Brent KJ, Holloman DW. 1998. Fungicide resistance: The assessment of risk. FRAC, Global Crop Protection Federation, Brussels, Monograph.;2:1-48.

[25] Costa TR, Fernandes FLF, Santos SC, Oliveria CMA, Liao LM, Ferri PH, Paulo JR, Ferreira HD, Sales BHN, Silva MRR.

[26] . Antifungal activity of volatile constituents of Eugenia dysenterica leaf oil. J Ethnopharmcol.;72:111-117.

[27] Fravel DR. Commercialization and implementation of biocontrol. Annu Rev Phytopathol. 2005;43:337-359.

[28] Fravel DR. Commercialization and implementation of biocontrol. Annu Rev Phytopathol. 2005;43:337-359.

[29] COSOVEANU A, AXÎNE O. and IACOMI B. 2010. Antifungal activity of macroalgae extracts. Scientific Papers, UASVM Bucharest, Series A, Vol. LIII, 2010, ISSN 1222-5339.

[30] El-ghanam AA, Farfour SA, Ragab SS (2015) Bio-Suppression of Strawberry Fruit Rot Disease Caused by Botrytis cinerea. J Plant Pathol Microbiol S3: 005

[31] Microalgae and wastewater treatment. Abdel-Raouf, N.; Al-Homaidan, A.A.; Ibraheem, I.B.M. (2012). Saudi Journal of Biological Sciences vol. 19 (3) p. 257-275

for their growth. Moreover, microalgae process can also be coupled as complementary process in secondary treatment (after bacterial processes). And also, their capacity to remove heavy metals, as well as some toxic organic compounds, therefore, does not lead to secondary pollution [32]. This field have been extensively studied. Lau et al. report the ability of Chlorella vulgaris in nutrients removal and reported a nutrient removal efficiency of 86% for inorganic N and 78% for inorganic P. In earlier study, reported an elimination of nitrogen (50.2%) and phosphorus (85.7%) in industrial wastewater treatment and elimination of phosphorus (97.8%) in domestic wastewater treated by microalgae.

Furthermore, emissions of CO_2 have become an important issue with respect to global climate change because atmospheric CO_2 concentrations increased significantly in the last century and have continued to rise at an increasing rate. As mentioned before (paragraph 1.4), biomitigation using microalgae represents one of the best options that merit to be prosecuted[33]. On the other hand, immobilization of microalgal cells is considered an efficient method to remove both nutrients and pollutions because it presents advantages with respect standard microalgal cultivation methods[34,35] . It will be interesting to confirm the feasibility of using immobilized microalgae and cyanobacteria for removing nitrate, ammonium, and phosphate from high volume effluent discharges. It has been reported that Phormidium laminosum immobilized on polymer foam has the potential to remove nitrate in a continuous-flow system with uptake efficiencies above 90%[36]. Indeed, major practical limitations of the two microalgal cultivation systems used today, shallow open ponds and tubular photoreactors, are the harvesting of the produced biomass. An efficient removal system is the key for recycling of wastewater.

The VALUEMAG immobilization method couple with magnetic cultivation is able to solve the drawbacks connected with biomass harvesting and CO_2 requesting. Despite the fact MAGMA cells need low quantity of water, there are no limitations on the quantity of MAGMA cells that can be growth on SOMAC surface. Furthermore, the thin layer of water that covers photosynthetic MAGMA cells warrantee an easy and efficient CO_2 capture without the use of bubbling systems (tubular phobioreactors) or mechanical mixing by paddle-wheel (shallow open ponds). Another strength point of VALUEMAG proposal is the versatility of MAGMA cells. Since any kind of microalgae can be transformed with SPANs it will be possible to stratify on SOMAC surface the best microalgae strain for each specific applications. In other terms, many microalgae present fast growing rate (double time <12 hours) but, e.g., low quantity of lipids, which implies that such microalgae strains are appropriate for fast CO_2 capturing (fast growth=high CO_2 fixation rate) rather than biodiesel production. **Approximately, half of the dry weight of microalgal biomass is carbon[37], which is typically derived from CO_2**. Thus, producing 1 kg of algal biomass fixes 1.6 – 1.8 kg of CO_2. Concerning water cleaning, based on the typology of pollution the best stains can be chosen. Table 4 resumes the most frequent organic and inorganic materials present in wastewater and the most appropriate microalgal species to remove them. Biological removal of nitrogen and phosphorus, for example, with microalgal is very efficient, many studies showed that more than 90% of these two elements are removed in less than 24 hours, however, as explained above the bottleneck remains the technologies that allow a fast uptake of the generated biomass. Since the biological performance of microalgae is already close to the maximum, VALUEMAG consortium will improved the technologies that will make affordable the employment of microalgae in wastewater schemes. VALUEMAG microalgae growing unit can be set nearby wastewater collector and operate independently by the other unit of the biorefinery.

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[32] A. Lau et al., 1996. P.S. Lau, N.F.Y. Tam, Y.S. Wong Wastewater nutrients removal by Chlorella vulgaris: optimization through acclimation. Environ. Technol., 17 (2) (1996), pp. 183–189.

[33] Wang, B., Li, Y., Wu, N., & Lan, C. Q. (2008). CO2 bio-mitigation using microalgae. Applied Microbiology and Biotechnology, 79, 707-718.

[34] de-Bashan, L. E., & Bashan, Y. (2010). Immobilized microalgae for removing pollutants: review of practical aspects. Bioresource technology, 101, 1611-1627.

[35] Ho, S. H., Li, P. J., Liu, C. C., & Chang, J. S. (2013). Bioprocess development on microalgae-based CO 2 fixation and bioethanol production using Scenedesmus obliquus CNW-N. Bioresource technology, 145, 142-149.

[36] S. Sawayama, K.K. Rao, D.O. Hall. Nitrate and phosphate ions removal from water by Phormidium laminosum immobilized on hollow fibres in a photobioreactor. Appl. Microbiol. Biotechnol., 49 (1998), pp. 463–468

[37] Mirón, A. S., Garcıa, M. C. C., Gómez, A. C., Camacho, F. G., Grima, E. M., & Chisti, Y. (2003). Shear stress tolerance and biochemical characterization of Phaeodactylum tricornutum in quasi steady-state continuous culture in outdoor photobioreactors. Biochemical Engineering Journal, 16, 287-297.

Nitrogen	*Chlorella vulgaris; Chlorella sorokiniana; Chlamydomonas reinhardtii; Isochrysis galbana; Scenedesmus obliquus*
Phosphorus	*Chlorella vulgaris; Chlorella sorokiniana; Chlorella emersonii*
Cadmium	*Chlorella homosphaera; Chlorella vulgaris; Chlamydomonas reinhardtii; Tetraselmis chuii*
Caesium	*Chlorella salina*
Chromium	*Chlorella vulgaris; Chlorella miniata; Scenedesmus acutus; Scenedesmu obliquus; Spirulina platensis; Anacytis nidulans; Chlorella sorokiniana*
Mercury	*Chlorella vulgaris; Chlorella emersonii; Chlamydomonas reinhard*
Uranium	*Chlorella regularis*
Mix of phenols, oil spill and heavy metals	*Chlorella sp.; Scenedesmus obliquus; Stichococcus sp.*

Table 1.3: Removal of pollutions by immobilized microalgae (adapted from ref. 32)

System integration by CO_2 recovery/sequestration and aqueous residual recycling aims to minimize the environmental impact of the VALUEMAG project. Micro-algae cultivation requires both carbon sources, such as CO_2, and nutrients for photosynthesis cycle and biomass growth and reproduction. Aqueous residual from biorefinery could be recycled and integrated into the micro-algae cultivation system as nutrient source[38] for algal growth, due to its composition, by carrying out a sustainable utilization of water resource. Before supplying, the degradation of potential inhibiting compounds has to be carried out. Several treatment can be used for water cleaning, such as chemical-physic processes, adsorption, filtration and oxidation. Due to CO_2 recovery, GHG emission from bio-refinery will be avoided, moreover micro-algae cultivation is an effective and environmental friendly way for a potential extensively sequestration of CO_2, in which 1.8 kg of CO_2 are fixed by producing 1 kg of micro-algae biomass[39]. Promising results to bio-mitigation of CO_2 emission, with CO_2 consumption rate in the range 200–1300 mg/L/day, were showed by several micro-algae strains, such as Chlorella sp., Scenedesmus sp., and Botryococcus braunii[38]. This way, VALUEMAG project offers the possibility to have the best recovery in terms of CO_2 as well as water reuse and could represent the best available technology for CO_2 sequestration from environment as showed below:

[38] Bjornsson W.J., Nicol R.W., Dickinson K.E., McGinn P.J., 2013. "Anaerobic digestates are useful nutrient sources for microalgae cultivation: functional coupling of energy and biomass production". Journal of Applied Phycology 25(5): 15231528. [39] Chisti Y. Biodiesel from microalgae. Biotechnol Adv 2007;25:294–306. [38] Lam M.K., Lee K.T., 2012. "Microalgae biofuels: A critical review of issues, problems and the way forward". Biotechnology Advances 30: 673–690.

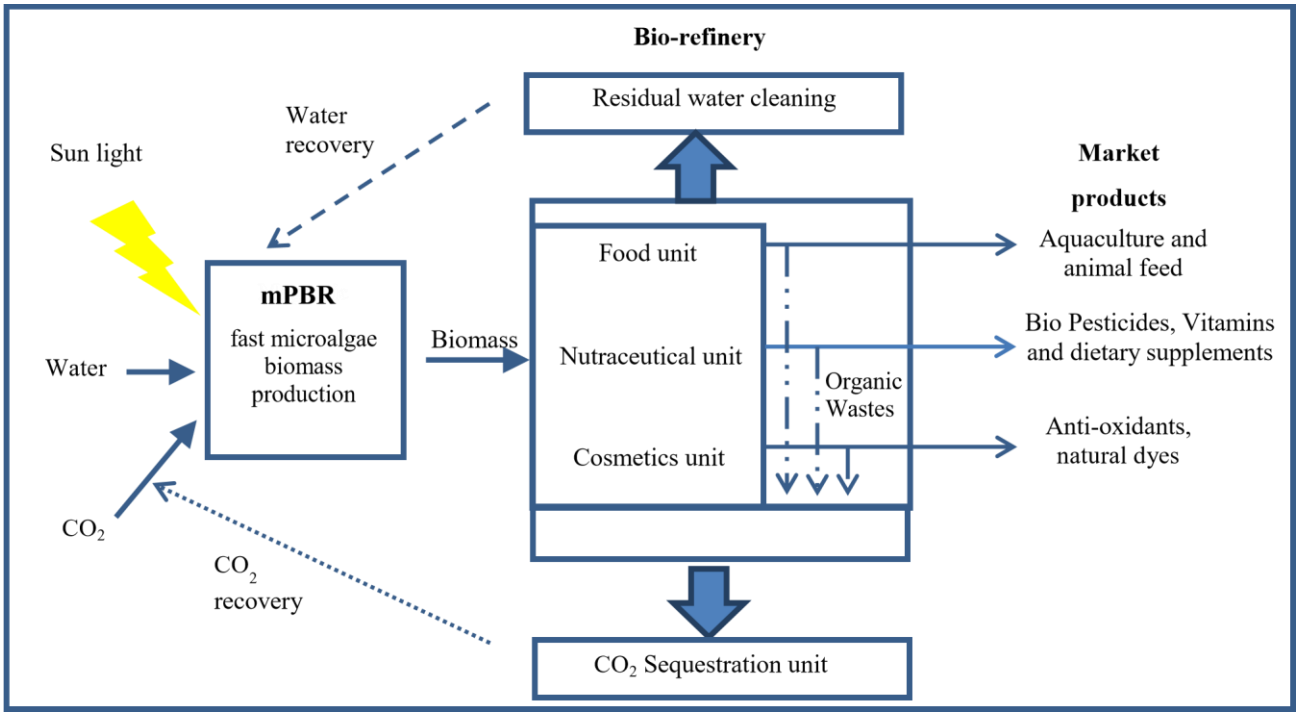


Figure 1.9:VALUEMAG magnetic photobioreactor (mPBR), biorefinery and CO_2 recovery/sequestration and water residual recycling

Selective Magnetic Separation of valuable compounds

In parallel to the well-established extraction methods, such as SFE, in the frame of VALUEMAG project it will be further developed a separation method based on the use magnetic particles (MPs) with a size >100 nm.

The innovative solution proposed to refine the purity of organic molecules extracted from biomass produced by VALUEMAG system is based on the use of MPs functionalized with surfactants and ligands able to specifically bind only one class of target molecule. The magnetic SEPARATOR is designed to process relatively small volumes of enriched solutions (>80%). It is suitable to further purify compound blend in order to reach a grade of purity >95-98% with a low running cost. The principle is based on the non-covalent binding of a specific organic molecule to a short ligand (e.g. oligopeptide, antibodies, metal ions and other chemical groups). However, by using MPs of different size it will be possible to use an additional scheme of separation based on different magnetic gradient. Indeed, MPs of different radius present different magnetic moment that can be used to selectively separate a specific compound.

The details of separation process are described below:

MPs functionalization: MPs of micrometer size are covered with a specific material that will allow to bind on them a short ligand, for example antibodies (Figure 1.10). The ligand has a particular conformation able to trap a specific target molecule, e.g. protein. The ligand composition is determined experimentally and is unique for a given specific compound.

MPs utilization : Functionalized MPs are spread on top of an electromagnetic surface in order to enhance the active trapping ability of them (Figure 1.11).

An electromagnetic surface is necessary since after MPs saturation (compunds absorption), they can be easily collected in a small vessel just by turning off the electromagnet. In this way no chemicals or mechanical methods are need to harvest MPs from the surface.

Separation of compounds from MPs: Since target molecules are trapped by ligand through non-covalent bonds the separation from MPs is done using mild chemical treatment or by changing physical parameters like pressure and

temperature. The treatment must be determined experimentally. However, after non-covalent bonds between ligand and target compound are broken, MPs are magnetically removed from the compound solution. Importantly, MPs could be re-used if the separation treatment do not damage the ligand structure.

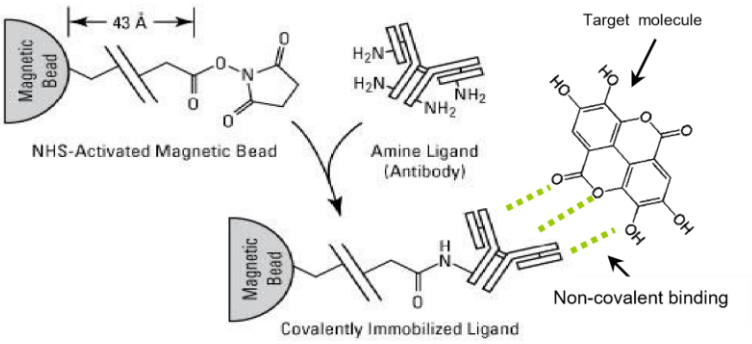


Figure 1.10: Functionalized MPs trapping target molecules through non-covalent bonds.

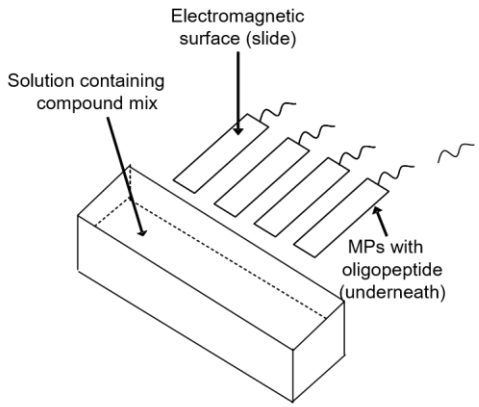


Figure 1.11: Each slide may contain MPs functionalized with different ligand in order to extract different compound at the same time

Life Cycle Analysis

Life Cycle Assessment for MAGMA and SOMAC micro-algae water trapping will be performed for food, cosmetic and nutraceutical production in order to define a self-sustainable system and to have an approach in accordance with the principles of the circular economy. In particular VALUEMAG project needs to be designed and operated in a way that ensures economic viability and that minimizes detrimental environmental impacts, at the same time.
Research lines in this field will focus on the following areas:

- Simplified LCA models applied to a wide range of microalgae sources, cosmetic, animal food and biopesticide utilization. These evaluations will take into account the particularities of locations where MAGMA plant may be deployed;
- Complete Life Cycle Inventory applied to a short selection of configuration plant technologies according to International Standard ISO14040. Operative objectives
- Producing simplified models of the Life Cycle Analysis (LCA) of alternative configurations and operating strategies of hybrid plants and quantitatively estimating the environmental impact associated with different stages of the energy generation cycle (including biomass growth and its uses) on different environmental categories;
- Evaluating the final costs for the technology scale-up of its coupling with cosmetics & animal food production as well as for bio-pesticides field;
- Producing a complete Life Cycle Inventory for a selection (2-3) of configuration plant technologies and operating strategies according to International Standard ISO14040, and feeding these data into a computational model in order to determine the environmental impact associated with the LCA of all the investigated configuration plants;
- Providing simplified assessments of the economic and energy efficiencies of alternative plant configurations and operating strategies.

System automation

A dedicated system monitoring and control system will be developed and installed in the newly designed mPBR and harvesting system created for the project. This system will contain the required sensing units to meet the requirements of the production of the microalgae to ensure the optimal growth conditions are met and the system is fully

autonomous for maximised production efficiency. Since microalgae are very sensitive to any medium variations or changes in the environmental conditions an intelligent learning system will be developed in order to predict and correct possible variations from the optimal growth conditions. To this end, the latest data intelligence tools will be applied to the system which will combine and model the data obtained by the sensors together with pre-loaded information from the environmental conditions (such as weather) or from the selected algae (such as optimal growth parameters from the literature).

- **System Integration**

The system integration includes hosting of all sub-systems inside the SOMAC (the soft magnetic cone) namely:

- The water system including MAGMA cells and the corresponding pumping system to lift the water flow on top of the SOMAC
- The SOMAC itself with the appropriate magnetization permanent magnets or electromagnets inside the cone and the required automation system for sequential excitation of different areas (rings) of the SOMAC
- The water circulation system including automation and computer control
- The de-watering system to concentrate microalgae
- The experimental apparatus to drive concentrate biomass (20-30% slurry) at the collection point
- Sensor systems to monitor all activities, including water flow, temperature, humidity of micrio-algae monolayer, magnetic properties of MAGNA cells

All these systems will be interconnected via a computer based control system to monitor all the above sub-systems and provide all required information to the researchers and involved partners. Such integration is schematically illustrated in Figure 1.12.

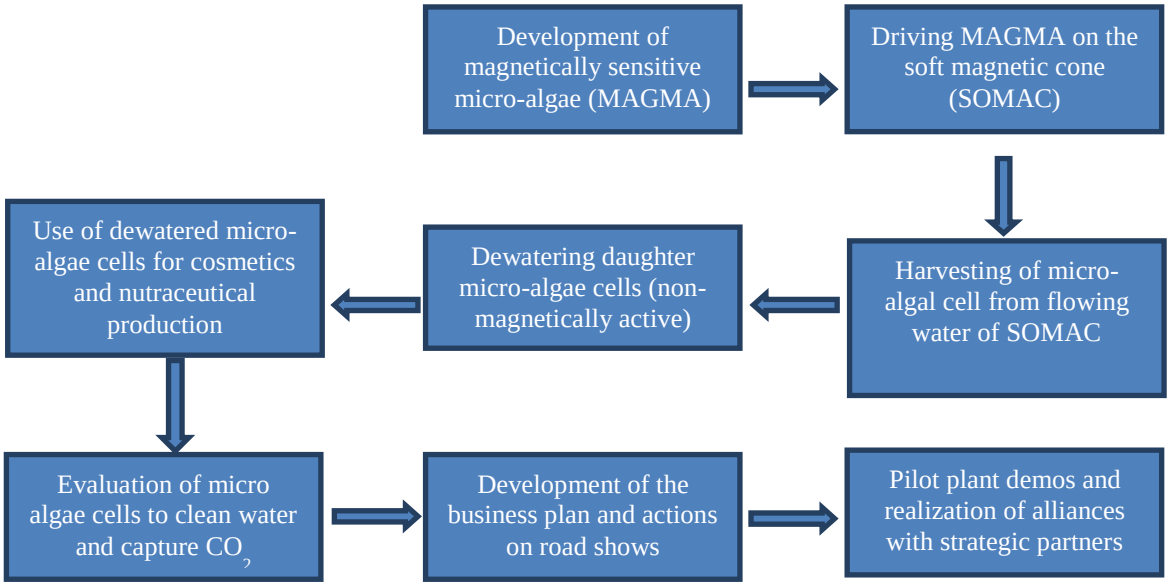


Figure 1.12: The system integration of the pilot plant of the algae cultivation

1.4. Ambition

Current state of the art and consequent motivation

The fundamental aim of VALUEMAG bio-refinery is to deliver high-value microalgae biomass-based products as a result of fast large-scale micro-algae cultivation. In order to attain this goal groundbreaking solutions in microalgae cultivation and biomass transformation are urgently need. Furthermore, it is also necessary to compensate

environmental issues connected with intensive land cultivation and global warming. Algae cultivation system like mPBR proposed in VALUEMAG project are new and innovative.

To our best knowledge, no pilot installations or state-of-the-art biorefineries utilize magnetic nanotechnology to cultivate microalgae. Biomass produced with mPBR will be directly utilized by a multi-facilities biorefinery for the production of food additives, nutraceutical, pharma and cosmetic products (Figure 1.9). **Multiple product extraction from algal biomass could be fundamental for the development of new strategies to produce highvalue outputs. It is unlike that micro-algae biomass production processes will be developed for, e.g., cosmetics as the sole end-production.**

VALUEMAG bio-refinery will provide the basis for multiple microalgae biomass component extraction and transformation by synergistically coupling different production processes. Furthermore, we believe that the production of high-value nutraceutical products from microalgae biomass can be the key to deal with the running cost of such technological advanced bio-refinery. In fact, since the magnetic immobilization is not species-specific many different type of microalgae, in separate SOMACs, can be cultivated even under the same greenhouse.

For example, astaxanthin, a natural human dietary component, has high commercial price (roughly Euros 4000 per kg) and its main natural source is the microalga *Haematococcus pluvialis*. It is estimated that the content of astaxanthin present in the microalga *Haematococcus pluvialis* is $40x10^3$ part per million (ppm), the highest ever detected in live organisms. Other example are the whole-cell dietary supplements that can be obtained from the cultivation of microalgae such as *Spirulla, Chlorella and Chlamydomonas*, which have a cost of Euros 50 per kg[40]. Cultivation of different kind of microalgae species, therefore, will guaranty an appropriated profit that will make sustainable the running cost of VALUEMAG biorefinery.

VALUEMAG project will also offer a fundamental step forward for the consolidation of technologies dedicated to the reduction of greenhouse gases such as CO_2. Climate change occurs mainly due to increased levels of CO_2 in the atmosphere. A number of research and development efforts have been directed at reducing CO_2 emissions and other kind of greenhouse gases. Among the most effective methods, biomitigation using microalgae is extremely promising. Indeed, decreasing of CO_2 emissions by fixing CO_2 through photosynthesis is very fast and efficient because microalgae have higher growth rates.

The ambition & innovation of the VALUEMAG project is related to:

- Dramatic reduction of the cost of algae production ○ Inexpensive production and take-up of magnetic nanoparticles required for algae cultivation ○ Inexpensive cultivation of algae based on our novel magnetic method ○ Inexpensive water removal from micro-algae
- Optimization of the use of the algae products ○ Production of animal food, bio-pesticides and cosmetics via CO_2-SF extraction technique ○ Production of animal food, bio-pesticides and cosmetics via our new selective magnetic separation ○ Cleaning up and recycling of residual water from bio-refinery and CO_2 recovery/sequestration
- Obtaining the proper alliances for the use of algae ○ Developing a business plan based on the lab results and the pilot plant operation ○ Determining strategic partner(s) to invest and fund the technology

○ Achieving sustainability in algae cultivation to support the action and other energy activities

Analytically, the reduction of the cost of algae production, as above mentioned is related to three main parameters, namely SPAN production, use of the SOMAC and implementation of harvesting.

- **Inexpensive production and taking-up of magnetic nanoparticles required for algae cultivation**

[40] Bergeron, C., Carrier, D. J., & Ramaswamy, S. (Eds.). (2012). Biorefinery co-products: phytochemicals, primary metabolites and value-added biomass processing (Vol. 19). John Wiley & Sons.

Figure 1.13. The CVD facility of NTUA, also facilitating for microwave assisted flame spray pyrolysis production of SPANs

Apart from that, the Slovak Academy of Sciences is uniquely equipped and globally well known for their characterization and modelling advances and ability. Their ultra-high resolution STEM facility, together with the multi-ability XRD machines, the DTA-DSC set-ups and finally the computer cluster able to perform ab-initio and multi-scale modelling, are able and sufficiently enough to characterize the produced. Figure 1.14, illustrates some of the results concerning magnetic nanoparticle characterization and modelling. With these particular results of modelling and comparison with TEM, XRD & DTA results, the total quality control of the produced nanoparticles is achieved.

Considering the small possibility of toxic effects of the SPAN on the micro-algae, although they have been proven to be not toxic for a period of 2-3 months, we offer the method and the tool to avoid such toxicity: covering SPAN with PEG, provides a safe and not-toxic environment for the micro-algae, where they are to be inserted. Complete covering of SPANs with PEG is an important issue. We have developed a method according to which a total quality control process has been developed to monitor the absence of holes in the PEG cover process. FTIR measurement, a non-expensive method of monitoring powders, can provide such information: Figure 1.14 illustrates the FTIR response with completely covered SPANs (solid lines in FTIR response – lower Figure), as well as the not complete response (dashed line at the left part As mentioned in the previous session, magnetic nanoparticles will be developed by microwave assisted flame spray pyrolysis: in other words we combine three top level technologies (vapour, flame spray and microwave energy) , the consortium will obtain the development of mass quantities of magnetic nanoparticles of any composition-stoichiometry, any diameter from 5 nm to 0.1 mm, covered by a plurality of shells. This technique, unique in the world will provide the ability of a single step production of PEG-covered SPANs.

Such a device exists in the Laboratory of the Electronic Sensors of the National TU of Athens. A small modification of the system, by adjusting an accurate gas precursor supply system of a cost of less than 20,000 Euros, together with a proper flame gun of several firing gases, in a cost less than another 20,000 Euros, the laboratory will have the ability of producing the required stoichiometry and size of nanoparticles, in mass of approximately 1 kg per day (lab scale) with size and stoichiometry uncertainty of 10 ppm. In Figure 1.13, one can see the set-up of thermal oxidation (CVD) facility of NTUA, which will also facilitate as the basis for the microwave assisted flame spray pyrolysis based production of mono-dispersed magnetic nanoparticles.

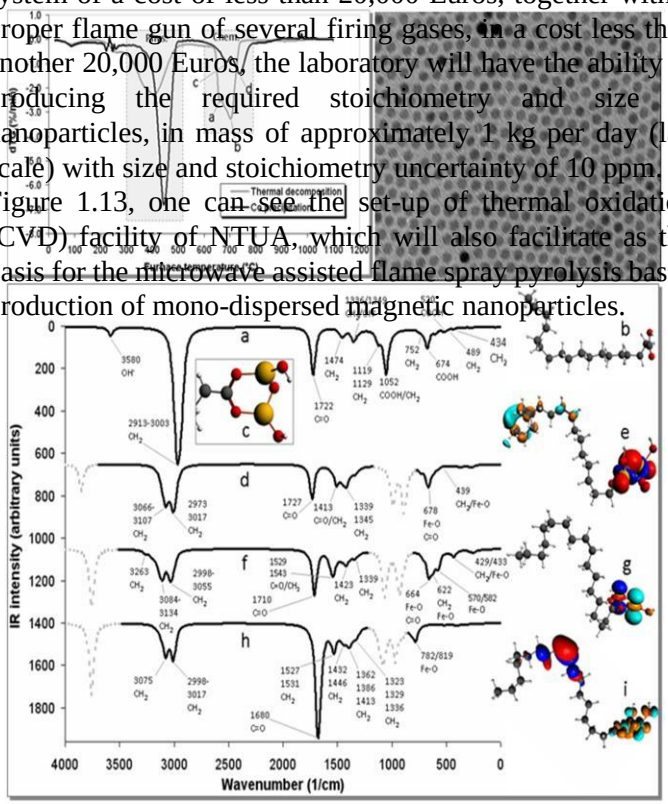
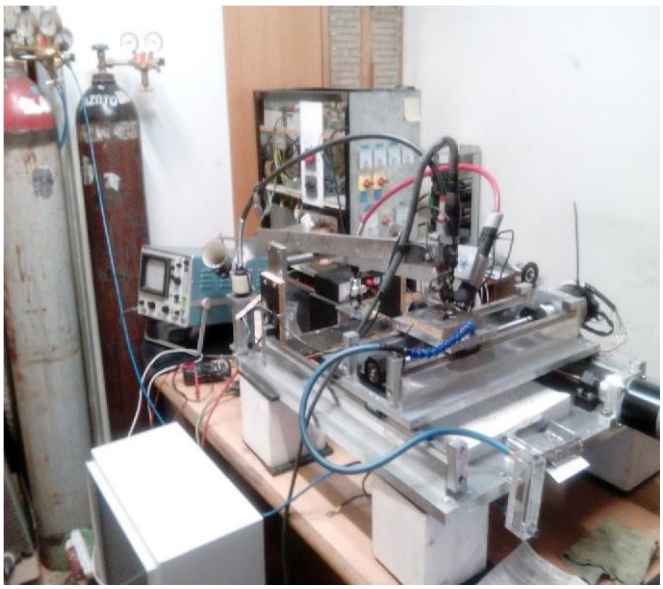


Figure 14. Initial results on magnetic nanoparticles characterization and modelling. Top left: DTA analysis of the thermal decomposition process to develop SPANs with flame pyrolysis. Top right: test of the production of SPANs (TEM image illustrating 5 nm monodispersed nanoparticles). Bottom: characterization and modelling of the SPANs.

of the lower FTIR response). This evidence is depicted in a publication of our group at the NTUA, warranting the total quality control of the SPAN manufacturing process.

Accordingly, the facilities of the workshop of the NOMASICO, assisted by NTUA, will be used for the electroporation of micro-algae.



The required additional cost in durables to obtain it is limited to 5,000 Euros in order to assembly existing systems used for electro-spark deposition of coatings on metals. A representative set-up of such system is illustrated in Figure 1.15. The whole system will be used to apply electric field in an industrial scale basin including algae in order to introduce magnetic nanoparticles into them.

Indeed, existing electroporation apparatus are available only at lab-scale. They can process each run few grams of cells in limited volumes (max 1ml). A good alternative is represented by the design of a flow-through electroporation machine that allows a continuous production of MAGMAs. The design and production of an industrial scale flow-through micro-algae electroporation **Figure 1.15. The basis for the electroporation system,** apparatus is therefore estimated in the order of less than **using the NOMASICO's electro-spark deposition method** 150,000 Euros, with a productivity of at least 10 kg per day.

- **Inexpensive cultivation of algae based on our novel magnetic method**

The most important challenge on the cost reduction of the algae cultivation process is based on the soft magnetic cone (SOMAC) methodology and operation. Bearing in mind that all types of algae need 400 nm – 700 nm electromagnetic radiation to be cultivated, the implementation of the SOMAC method in suitable warehouse in the Mediterranean Sea is of absolute importance and efficiency:

- The direct exhibition of all algae on the Mediterranean sunlight due to the magnetic method of trapping and maintaining their position on the magnetic cone, warrantees their proper cultivation – multiplication
- The implementation of the cultivation chamber as the proper environment of the SOMAC, warrantees avoiding problems of contamination and optimization of algae exposure to sunlight in more or less constant conditions of humidity and temperature

Estimation of biomass production is based on the following assumptions:

- Single-cell algae are considered spherical and with the same density of water
- Single-cell algal diameter: 5 μm
- Surface occupancy of a single algal cell: 25 μm^2 (5 μm x 5 μm)
- Volume of a single-cell alga 65 μm^3
- Weight of a single-cell alga 65 pg
- In 1 m^2 of magnetic surface can be spread ($1/25 \times 10^{-12}$) 4^{10} algal cells
- Biomass that can be spread on 1 m^2 is 2.6 grams (4^{10} cells x 65 pg)
- By increasing the magnetic surface (roughness) of a factor equal 4, in 1 m^2 can be spread roughly 10 grams of microalgae.
- Thickness of water covering cell monolayer 0,05 – 0,1 mm

Thanks to technological improvements achieved using mPBR, the growth rate of microalgal cells can reach the maxim theoretical level; in fact each single cell will receive the same amount of light fluence, gas exchange and temperature will be constant all over the cell monolayer; whereas the constant mixing coupled with harvesting will ensure that no culture growth saturation (plateau phase) will be reached. These condition can guarantee a quasisynchronized cell growth that, by using the appropriate microalgal species, can lead to double or even triplicate the amount of parental cells within a day. Assuming that on top of 10,000 m^2 (1 hectare) of magnetic surface can be spred 100 kg of MAGMA, each day the production will be at least 100 kg (pretentious doubling) of pure dewatered

biomass. It must be highlighted that 1 hectare of magnetic surface does not need to be place on the same land-surface area. In fact, first the mPBR are designed to be developed vertically (several meters high) with particular shapes such as pyramids or inverted cones; second a unique magnetic surface fabrication process will allow to expand (roughness) the magnetic surface. In this way 1 hectare of magnetic surface, reduced by 4-5 orders, will lay on a land-surface area of 2000 m^2. Furthermore, the amount of water used in mPBR is dramatically reduced (1 litre/m^2 assuming a 1 mm thickness film of water covering the microalgae monolayer) with respect the actual average amount of 20-30 litre/m^2 used in raceway open ponds.

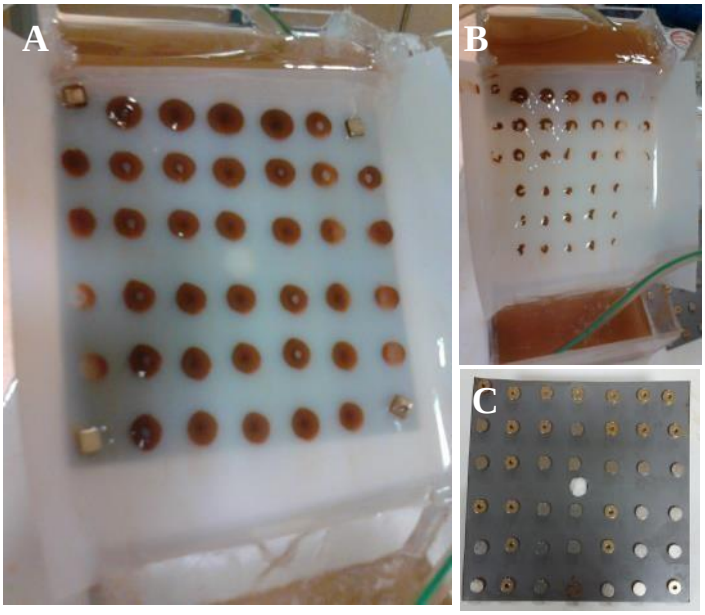


Figure 1.16: Magnetic surface prototype. A) SPANs are immobilized by an array of small neodymium magnets and covered by a tiny film of circulating liquid. B) As the magnets array is removed, SPANs are wiped off by the water flow. C) Array of small neodymium magnets used to magnetize a plastic surface.

This is confirmed by a preliminary prototype (Figure 1.16), built at NTUA, of an inclined magnetized surface on which SPANs are trapped by an array of small magnets, while water gently falls down by gravity. The prototype proves that a thin layer of water can flow and cover SPANs without wash off them, in fact by removing the magnets array SPANs are dragged down by the water flow.

With this new technique when MAGMA cells are magnetically immobilized the concentration of the generated microalgae cells will be much higher with respect to actually existing technologies. In fact, though variable, with standard system concentration in the order of 1-2 mg/l, whereas with the VALUEMAG-mPBR it can reach 3-5 g/l allowing to harvest biomass with one-step process. Bio-refineries based on this new biotechnological advanced mPBR can provide biomass with unprecedented quality and quantity.

With the expected yield both the volumetric productivity and the areal productivity of microalgae biomass will be extraordinarily increased as compared with the actual methods. Indeed, a recent cost-analysis study focused on open ponds, horizontal tubular photobioreactors and flat panel photobioreactors, showed that the costs of microalgae biomass production including dewatering, were 4.95, 4.15 and 5.96 € per kg, respectively[41]. The important cost factors considered were irradiation conditions, mixing, photosynthetic efficiency of systems, medium and carbon dioxide costs. Another study showed that the cost of photobioreactors is the major factor in the production cost[42]. In fact, if that could be reduced while maintaining the productivity, the production cost would decrease. Furthermore, the next factor in relevance is the consumption of raw materials. Carbon dioxide is the most expensive consumable, the use flue gases from industrial sources can reduce the cost of CO_2 to values as low as zero if flue gases are readily available. Finally, the uptake of water and biomass harvesting also contributes significantly to the production cost. Accordingly, it has been estimated that by optimizing all the above mentioned parameters, the unit production cost can be reduced for tubular or flat panel photobioreactors to 0.70 and 0.68 €/kg, respectively, whereas for open raceways the cost cannot be reduced below 1.28 €/kg. The innovative cultivation method proposed by VALUEMAG consortium will allow to further reduce the cost (□0.3 €/kg) of biomass production with respect the tubular or flat panel photobioreactors. Indeed, unlike these systems, no expensive infrastructure must be purchased, CO_2 produced by transformation processes

[41] Norsker, N. H., Barbo sa, M. J., Vermuë, M. H., & Wijffels, R. H. (2011). Microalgal production—a close look at the economics. Biotechnology Advances, 29, 24-27.

[42] Acién, F. G., Fernández, J. M., Magán, J. J., & Molina, E. (2012). Production cost of a real microalgae production plant and strategies to reduce it. Biotechnology advances, 30, 1344-1353.

done inside VALUEMAG plant can be used for microalgae growth, the volume of water is very low considering that cells are covered by a thin flow liquid and harvesting cost are reduced thanks to the high biomass concentration.

- **Cosmetics, Animal Food and nutraceutical**

According to the composition of cosmetics, several micro-algae will be selected and cultivated; in particular, Spirulina and Haematococcus pluvialis micro-algae will be used. They will be characterized in terms of Astaxanthin content, which is a natural anti-age with antioxidant power 550 times higher than E-vitamin and 11 times higher than β-carotene.

A Supercritical Fluid Extraction (SFE) based process, developed by ENEA, can be used for the Astaxanthin extraction. According to this technique, a simpler procedure is needed in comparison to the traditional extraction methods. Moreover, organic solvents are not required since a gas in supercritical conditions is used as solvent**. A double advantage is achievable by using a supercritical gas**: on one hand, high diffusivity and low viscosity, typical characteristics of a gas, are preserved; on the other hand, high density and low compressibility, typical of a solvent in liquid state, are obtained. These characteristics allow the use of SFs as solvents for organic molecules. Furthermore SFE process has high extractive yield and low extraction time, at the same time. **For VALUEMAG project CO_2 will be used as supercritical fluid since critical conditions (T = 31°C and P = 74 bar) are easily achievable.** The low critical temperature of CO_2 reduces the degradation of thermos-labile molecules. Furthermore, the presence of impurities among the extracted molecules is avoided thanks to the total recovery of gaseous CO_2, during the separation between CO_2 and extracted molecules. The same process will be used for the extraction of Polyunsaturated Fatty Acids (PUFA), like lipids that are used in animal diets. PUFA perform metabolic functions and avoid inflammatory processes in animals.

Experimental tests will be carried out using bench scale CO_2 SFE plant for a first screening of microalgae (Figures 1.17, 1.18), in particular Spirulina and Haematococcus pluvialis at the end to obtain the best performance of process in term of yield and purity for the two main bioproducts usable in Cosmetics & Animal Food fields as Astaxanthin and PUFA. After this step the best performance obtained in bench scale will be replicated in a CO_2 SFE pilot scale and each bioproduct will be analysed in the laboratory in both terms of quality and quantity. Astaxanthin will be analysed using SFE analyser as soon as high-performance liquid chromatography (HPLC) on normal phase associated with detection of the visible spectrum (NP-HPLC-VIS, 470 nm) according to EU 2015/1415 regulation for Astaxanthin uses. At the same time PUFA will be analysed via Gas chromatographic determination of total fatty acids, saturated and unsaturated fatty acid and sterols.

Figure 1.17: CO_2 SFE bench scale plant – volume: 1 litre

Figure 1.18: CO_2 SFE pilot scale plant – volume: 15 litre

Figure 1.19: Spray-dryer for lyophilization (Capacity: 20kg/h)

In order to concentrate PUFA and Astaxanthin, a freeze-drying system will be used (Figure 1.19). This technology allows water removal from an organic substance minimizing the deterioration of structure and components. By this way a good storing at room temperature, with minimum structural changes, will be obtained and odour, flavour, colour and nutrients will be abundantly preserved, at the same time.

Downstream processing is a major issue in microalgal biotechnology. Firstly, the biomass needs to be separated from liquid media, which is usually done by centrifugation, although several alternative methods can be used, such as flocculation, filtration, etc.. In VALUEMAG proposal the SFE CO2 technique for downstream processing of high added value compounds by microalgae will be used. This method, whose characteristic is the cleanness and the lack of toxicity of the extracted compounds thanks to the use of CO2 as a solvent, will be used for the recovery of high value pharmaceutical/nutraceutical substances. The research work will define the best operative conditions and will produce interesting findings for the relevant industry. The drawback is that in the SFE CO2 process the extraction efficiency is not so high as in classical solvent extraction. In order to improve the efficiency in VALUEMAG the presence of an optimum co-solvent will be tested and, for each extracted molecule, the parameter process will be optimized. The equipment cost for SFE CO2 extraction is higher than that of classical solvent extraction. A comparison between SFE CO2 and traditional solvent extraction is shown below:

Traditional solvent extraction	SFE CO_2 extraction
Presence of solvent is inevitable; residual solvent concentration (usually in the order of ppm) depending on the solvent used	Procedure is completely free of solvents and thus extracted compounds are pure
Heavy metal contamination is also unavoidable, and depends on the solvent recycling procedure, source of raw material and what the machinery parts are made from	Free of heavy metals; which are not extracted, even if they are present in the raw material.
Inorganic salt content is also difficult to avoid	Free of inorganic salts
Solvents have poor selectivity; during solvent removal, polar substances form polymers which lead to discoloration of the extract and poor flow characteristics	CO_2 is highly selective, consequently there is no change of polar substances forming polymers
Both polar and non-polar colors are extracted	Only non-polar colors are extracted
Solvent removal requires extra unit operations, which results in higher cost and lower recoveries	No extra unit operations is required, and yield is very high

Table 1.4 - Comparison of solvent extraction and SFE CO_2[43]

By using SFE for the extraction of PUFA, Carotenoids and other high value molecules in Valuemag it will be possible to increase the yield by 10-15% with respect to the extraction with traditional liquid solvents that a is of about 80%, with lower process costs as compared to the traditional ones. The main limits on the SFE CO_2 process is the capital costs related to plant construction. VALUEMAG will assess the size of SFE CO_2 process necessary for obtaining an average cost, for biobased products, lower than 5euro/kg. This cost is, in turn, 20% lower than the conventional ones. This result will be reached also thanks to the addition of co-solvent to increase the extraction efficiency. The extracts obtained by SFE are highly concentrated since the solvent is separated by process depressurization making the extract free of residues. Moreover, in SFE CO_2 process the resulting CO_2 gas stream can be recycled, making SC-CO_2 extraction an environmental-friendly process. Efforts should focus on the reduction of product loss and equipment and energy costs associated with the extraction and purification steps. **Life cycle assessment**

VERTECH will apply their competences in LCA for detailed quantitative environmental sustainability evaluation of magnetically enhanced micro-algae cultivation analysed within the frame of the VALUEMAG.

The environmental impact assessment will be conducted using the Life Cycle Assessment (LCA) methodology in compliance with ISO 14040 and 14044. In the first step, goal and scope of the specific system will be jointly defined with the consortium partners. The functional unit will be defined in order to enable a scientific comparison of novel technology alternatives. Secondly, a detailed life cycle inventory data on energy consumption, material use, and emissions, will be established for the magnetic micro-algae cultivation based on information gathered within

[43] Mercer, P., & Armenta, R. E. (2011). Developments in oil extraction from microalgae. European journal of lipid science and technology, 539-547

VALUEMAG as well as from literature. The life cycle inventory models will be set up in the LCA software SimaPro 8 and connected to background inventory data from the Ecoinvent v3 database.

For the impact assessment, a broad range of life cycle impact assessment methods will be applied and the respective indicators will be calculated, such as greenhouse gas emissions according to IPCC 2013 cumulative energy demand, etc. Based on a contribution analysis, environmental hotspots will be identified and recommendations for ecological technology improvement (Eco design) will be developed.

The results will allow decision-makers to include environmentally relevant aspects in their decision process and most importantly to demonstrate the advantage in terms of increased sustainability of magnetically enhanced cultivation of micro-algae.

- **System automation**

Many aspects of the mPBR will be taken into account in the project in order to maximise the yield in biomass production. These will need to be monitored and controlled through a dedicated system focussing on the various parameters of interest for the process such as the CO_2 supply system, light source, O_2 removal system, pH and temperature control, harvesting system, among others. The VALUEMAG system will automate the care and control of the optimal conditions for microalgae growth and production.

An automatic and easy to use control system will be designed for the VALUEMAG mPBR, which will be capable of controlling and adjusting a photosynthetic microorganism culture in order to arrive at optimal operating parameters in keeping with variations in different wastewater and associated processes. Biological systems do not possess a linear reaction to external parameters, but often have several relative production maximums which pose operational challenges. To this end, the operating point must be stable against foreseen operating changes. In addition, the memory of biological systems may lead two systems operating under the same conditions to behave differently. To overcome these limitations, the VALUEMAG control system will be based on fuzzy logic (or similar data mining techniques) and PID and will allow the mPBR operator to automatically select the optimum parameters of growth depending of the microorganisms and additional conditions. In addition, the PBR will be equipped with various sensors (light in the suitable range to allow sufficient photosynthesis without leading to saturation, suitable concentration in CO_2, O_2, adequate pH and temperature, etc.), and will provide an automatic assessment of the resulting yield of the biomass by turbidimetric measurements. When yield is sufficient a harvesting system will recover the biomass.

Considering that open pond cultures are the most economical way to produce microalgae, the control system designed and built in this project will approximate that of an open pond facility which will enhance the future scale-up. Depending on the micro-algal species, various harvesting processes exist for photobioreactors, gravity filtration, microstrainers, lamella separators, belt filters, vacuum drum-filters and chamber filters press with the most common being flocculation and centrifugation . Based on this, the developed control system will have to adapt to the different types of harvesting systems in order to allow for an easy location of the sensors and monitoring devices while ensuring a smooth operation of the harvesting techniques. In addition, VALUEMAG will tap into the ability of photosynthetic microorganisms to use CO_2 as a carbon source converting it to organic carbon and incorporating it to their biomass.

- **Obtaining the proper alliances for the use of algae**

The final important ambition of the VALUEMAG project is related to the proper organization and approach for the alliances with proper organizations.

The first step in this challenge is related to the realization of the business plan: the results of business plan operation will offer all necessary data to provide a trustful business plan and the sustainability parameters of the method for algae cultivation as well as for producing valuable products out of the algae cells.

The second step is (mainly based on the business plan) the decision to demonstrate the results of pilot plant operation, mainly on the cost and quality valuable products, such as cosmetics, food and environmental activities and pharma molecules (proper choose of expo-shows)

The third step is related to the alliance with given entities and/or organizations. The collaboration of the consortium with Classification Societies, such as the DNV-GL, the largest Classification Society in Europe, permits access for presentation and auditing of the method. Furthermore, the VALUEMAG consortium will react with the local coastal

societies of Cyprus, where the pilot plant is to be developed, in order to persuade them about the validity of the project and the corresponding economic importance for them.

Strengths	Opportunities
The main strength of the proposal is related to the significant decrease of the cost of micro-algae production. The proposed magnetic cultivation method promises, due to the immobilization of micro-algae, to the sunlight exposure, and the natural way of flowing a thin layer of water on top of them, high productivity and easy harvesting of a large variety of micro-algal strains. The other proposal strengths are the ability to produce high quality and valuable cosmetics, food and nutraceuticals, as well as to provide sustainable ways of water cleaning and CO_2 capture. The holistic approach of the process enables a circular process with zero wastes and a high valuable final products.	The main opportunity in that the current economy is related to the penetration in the biomass production market. The important opportunity is the ability to produce high quality cosmetics, food and nutraceuticals, open new markets in different directions. Apart from that the need of clean water in many areas of the planet, together with the necessity for CO_2 footprint reduction is also an important issue. Finally, a new technology of algal cultivation will be provided to European industries along with a new integrated biorefinery concept allowing for the possibility of usign micro-algae cultivation for this purpose.
Threats	**Weaknesses**
The main threat comes from the pharma & food additives industry. The main issue comes from the reduction of cost of such high-value end products, such as the micro-algae primary and secondary metabolites. There are several ways of "killing" the principal idea, by means of running counteracting research and development to prove the non-validity of these results. Road shows are the solution to counteract on these disputes. The other way is to ally with the inventors in order to delay such production possibility. The way to counteract on that is the actual interest of the inventors in shares and options of the product when it is ready to be in the market.	The main weakness of the whole attempt and the technological method is the need of production in isolated areas for practical reasons (exposure to sunlight, proper conditions of air to permit proper Cultivation chamber activity etc.). This way, it is unattractive for young people to change life in order to follow a new opportunity of production. The way to counteract on that problem is the selection of areas of micro-algae cultivation, combining proper climate and some town-like activities. Such an example is Cyprus and its related environment-climate, where the consortium of the VALUEMAG project decided to develop the pilot plant of the project. Other weakness of the VALUEMAG project is to be the first project applying this concept, despite of the number of publications and papers already available

Figure 1.20: SWOT Analysis of the proposed Tech

2. Impact

2.1. Expected Impact

If we count all economic activities that depend on the sea, then the EU's blue economy represents 5.4 million jobs and a gross added value of just under €500 billion per year. Much of this activity is concentrated around Europe's coasts. The sea and the coasts are drivers of the economy and the aquatic-biomass bio-refineries are a new dynamic field in this blue economy.

VALUEMAG provides an effective answer to demonstrate and scale up the pilot production and harvesting of integrated algae products and bring them nearer to the market in an economically, environmentally and socially sustainable manner, specifically addressing the topic BBI-2016-R09 "Exploiting algae and other aquatic biomass for production of molecules for pharma, nutraceuticals, food additives and cosmetic applications".

In terms of the expected impacts that are listed in the BBI JU annual Work Plan as well as in the work program, VALUEMAG will contribute as follows:

<u>Contribution to the expected impacts set out in the BBI JU annual Work Plan</u>

- ***Achieved new process to separate, purify and/or extract the compounds of interest at a sufficiently high yield and low cost to allow a cost-effective scale-up of the process.***
- ***Achieved improved yield per mass unit biomass-input, or improved cost effectiveness as compared to current routes towards same or comparable products.***
- ***Contribution to the BBI JU Key Performance Indicators (KPI), specifically: Set the basis for the realization of 1 new bio-based value chain (KPI 2), new bio-based materials (KPI 5) and new demonstrated consumer products based on bio-based chemicals and materials (KPI 6), or significantly improving the production process of existing ones.***

The VALUEMAG project will develop an innovative technology able to cultivate algae in a minimum possible cost, much lower than the existing cost of micro-algae cultivation and will integrate this with other novel technologies in a pilot-scale bio-refinery. The new integrated production system will have the potential to increase the efficiency and the environmental sustainability of modern bio-refineries through lowering the cost of securing cleaner energy and sustain food security.

Thanks to technological improvements achieved using mPBR, the growth rate of microalgal cells can reach the maximum theoretical level; in fact each single cell will receive the same amount of light fluence, gas exchange and temperature will be constant all over the cell monolayer; whereas the constant mixing coupled with harvesting will ensure that no culture growth saturation (plateau phase) will be reached.

Conditions based on the previous mentioned estimations can guarantee a quasi-synchronized cell growth that, by using the appropriate microalgal species, can lead to double or even triplicate the amount of parental cells within a day. **Assuming that on top of 10,000 m^2 (1 hectare) of magnetic surface can be spread 100 kg of MAGMA cells, each day the production will be at least 100 kg** (pretentious doubling) **of pure de-watered biomass**.

It must be highlighted that 1 hectare of magnetic surface does not need to be placed on the same land-surface area. In fact, first the mPBR chambers are designed to be developed vertically (several meters high) with conical shapes; second our unique magnetic surface fabrication process providing the necessary magnetic field gradient plays another role too: it allows for expanding the magnetic surface to host the MAGMA due to its roughness. In this way, one hectare of magnetic surface reduced by 5 times of magnitude at least, due to the surface roughness of the SOMAC (soft magnetic cone), results on a land-surface area of 2,000 m^2.

Additionally to that, the conical arrangement of the SOMAC, offering a cone of 6 m height and 6 m diameter, thus offering an exposed surface of 63,2 m^2, where the corresponding surface of the land equals 28,2 m^2, the actually required land is reduced down to half. Therefore, the total land required for 10,000 m^2 of cultivating mPBR area corresponds to a land equal to 1,000 m^2 including the gained surface due to both the conical geometry as well as the mentioned roughness: the first contributes about double the surface, while SOMAC roughness contributes to 5 times more surface, except the advanced ability of magnetic trapping due to the roughness-induced magnetic gradient, responsible for increasing the magnetic forces exerted between MAGMA and SOMAC

Thus, 1,000 m^2 of SOMAC surface can offer an effective surface of 10,000 m^2 (1,000 X 2 X 5) producing 100 kg of pure de-watered biomass per day or more than 35 tons per year. This figure of merit suggests a great improvement with respect to the existing state of the art in bio-mass production, since it was estimated that micro-algae biomass production ranges between 15 and 25 tonne/ha/year[44].

An initial cost estimation of algae-based biomass production with the new technique has been made. It can be assumed that certain cost will decrease when the production site would be scaled up to larger areas. This has an effect on capital good costs, labour requirement and extra land requirement for the surrounding equipment and infrastructure. In estimations the scaling parameter was considered 0.6.

- **Micro-algae raw material.** The cost using open ponds, horizontal tubular photobioreactors and flat panel photobioreactors, to produce microalgae biomass are 4.95, 4.15 and 5.96 € per kg, respectively. The innovative cultivation method proposed by VALUEMAG consortium will allow to further reduce the cost

[44] M.K. Lam, K.T. Lee, 2012. Microalgae biofuels: A critical review of issues, problems and the way forward. Biotechnology Advances, 30: 673–690.

of biomass production up to ~0.3 € per kg (see: *Inexpensive cultivation of algae based on our novel magnetic method*, page 23). This is due to the energy-inexpensive and fast process of MAGMA development.

- **Cultivating water.** The amount of water used in mPBR is dramatically reduced (0,1 litre/m^2 assuming 0,1 mm maximum thickness film of water covering the microalgae monolayer) with respect the actual average amount of 20-30 litre/m^2 used in raceway open ponds. Thus, 10,000 m^2 of cultivating MAGMA area (corresponding to 1,000 m^2 actual land covered by SOMACs) require only 1 ton of water. This implies that concentration of microalgae cells will be much higher with respect to actually existing technologies of the current state of the art. In fact, though variable, considering a standard system concentration in the order of 1-2 mg/l, whereas with the mPBR it can reach 2-3 g/l, allowing for biomass harvesting implementing onestep process.
- **Energy.** The energy required is minimized to climate conditions inside the warehouse, pumping water, dewatering process and packaging. It is roughly estimated that 30 kW of instant energy is required for 1,000 m^2, or 750 kWh per day, or 75 Euros per day (at the most expensive electricity rate), or 3,000 Euros per year.
- **Labour.** The required personnel for a micro-algae farm concerning automated large scale farms, is roughly estimated to 10 technicians per 100,000 m^2, including all activities, like maintenance, or 0.1 technician per 1,000 m^2, or ~2,000 Euros per year. Bearing in mind that the total cost of labour and consumables is less than 10,000 Euros per year per 1,000 m^2 of land, offering more than 35 tons of bio-mass.

Thus, the innovative VALUEMAG microalgae cultivation-production process can provide sustainability in algae production, due to the low cost and fast process in comparison with any other state-of-the-art process.

Apart from that, the implementation of both valuable-product extraction techniques, the supercritical CO_2 process and the selective magnetic separation methods, or their combination, will provide additional benefits in the quality of the produced results.

The feasibility analysis of those processes corresponds to certain Tasks of the VALUEMAG project. At this stage it is not as easy as for the case of the micro-algae cultivation process to offer a rough estimation on the cost of the valuable products, extractable from the micro-algae.

- ***Safety, quality and purity of the (new) products are in line with EU legislation and proven to meet end market requirements in order to facilitate future market access and commercialization***

The safety in producing the micro-algae and the corresponding high-value products is substantially increased with respect to the existing state of the art in terms of safety of the working people, as well as the safety of the property invested: the proposed technology cannot cause any harm to employees of the algae farm. Apart from that, the proposed technology cannot cause any damage in the invested property: the methods of production have been proven in a preliminary stage and will be fully proven during the VALUEMAG project to be pretty harmless for the infrastructure of the algae parks.

The quality and purity of the new products, either in terms of the cultivated-produced dewatered micro-algae cells or the valuable products extracted from those micro-algae is significantly higher than the current state of the art:

- The selective magnetic separation of the NTUA warranties completely pure extraction of all ingredients due to the unique correlation of ligand and micro-algae ingredient: it is impossible for a ligand able to trap a given micro-algae ingredient, to trap another kind of ingredient.
- Furthermore, the use of the super-critical CO2 extraction method, where ENEA is an expert, warranties the proper extraction of high quality and purity high-value ingredients from the micro-algae cells.
- Finally, the possible combination of the two above mentioned methods may also provide significant additional promising results, allowing for an even better cultivation and treatment process.

The sustainability of production is a topic that concerns the entire value-chain. VALUEMAG will give special focus on environmental sustainability by performing Life Cycle Assessment (LCA) (WP4 -Task 4.6) for the overall environmental impact of the new technology and products for a pilot plant, as well as large scale application for the

entire chain. The evaluation will take into consideration direct or indirect land use change (LUC and iLUC) as well as quantification of the impacts of production on regional biodiversity.

The VALUEMAG within the frame of specific activities (WP3-Task 3.7) will examine approaches like risk management and certification schemes for good and sustainable practices, such as Eco-Management and Audit Schemes. VALUEMAG team will examine the biosafety in two dimensions: (a) potential adverse environmental consequences from the processes and (b) potential harm to human or animal health in case of food/ feed or pharmaceutical applications. For human health risks, the biosafety evaluation will be based on methods applied in higher plants to guarantee that they are safe and they do not produce toxic substances or allergens. The environmental risks assessment will be based on well-known risk analysis procedures.

Furthermore, an assessment of regulatory and policy framework regarding the proposed process and the final products will be performed (WP4- Task 4.7). The assessment will cover whole production chain (research, production, and commercialization) and will involve a survey/study on EU and international regulations affecting the VALUEMAG plants and products including among others: EC directives 2009/41/EC, EC directive 2001/18/EC , EC directives 2009/41/EC, EC directive 2001/18/EC, EC Regulation on Food Safety (EC 178/2002), EC Regulation on Novel Foods and Novel Food Ingredients (EC 258/97), EC Regulation on Nutrition and Health claims made on foods (EC 1924/2006), EC Regulation on traceability and labelling of GMOs and of food and feed products produced from GMOs (EC 1830/2003) etc.

All the above actions will have great impact on the safety, quality and purity of the delivered products always in line with EU legislation and market requirements.

Contribution to the innovation capacity and the integration of new knowledge, strengthening the competitiveness and growth of companies

The European Union has adopted an ambitious strategy for developing the Bio-economy in Europe. The Bioeconomy is based on the innovative use of sustainable biological resources to cover the growing demand of the food, energy and industrial sectors. In this context, algae and other aquatic biomass represents an emerging biological resource of great importance for its potential applications in different fields, including food and feed.

The proposed pilot-scale bio-refinery plant offers the opportunity to revitalize rural and marine Europe by creating a new manufacturing base, linking R&D activities to local/regional development, maximizing the social and economic benefits. VALUEMAG partners recognize that research on algae and other aquatic biomass bio-refineries is not only a technological issue, but as much an economic, political, environmental, food-security and energysecurity one. Being a cross-cutting subject, the consortium will promote interdisciplinary topics, will integrate a range of key actors and will reach knowledge excellence.

The majority of the novel and innovative contributions of the VALUEMAG project come from the involved Universities and Research Centers, like NTUA, SUN, ENEA and FU SAV:

- NTUA is providing novel and innovative technology on algae cultivation as well as on selective magnetic separation of valuable products from micro-algae cells
- SUN & ENEA are providing novel and innovative technology of super-critical CO_2 based production of high-added value products from the micro-algae cells
- FU SAV is providing their valuable know-how in structural characterization of micro-algae and the resulting valuable products, correlated necessarily with their properties

Nevertheless, the final result, namely the actual plant to cultivate micro-algae cells and extract their valuable ingredients wouldn't be possible without the sincere contribution of the involved SMEs:

- EPARELLA provides de-watering systems and knowledge about micro-algae biology
- NOMASICO provides the greenhouse and the development of the SOMAC
- IRIS provides the automation systems and the integration of the pilot plant in an automated system

- THERACELL provides its know-how in the commercialization process of those high-added value products
- EXERGY provides the required modeling of the whole plant and all operations-processes
- VERTECH provides the life-cycle analysis and the feasibility of the project
- PNO provides the dissemination of the results of the pilot plant for the benefit of all partners

This way, it is clear all enterprises involved in the VALUEMAG project have their discrete role, without missing any kind of activity and therefore allowing for a concrete consortium of SMEs able to unify forces and power to provide the new age in micro-algae cultivation. For this reason, the business plan of this activity is a crucial Task, predicted in the VALUEMAG project.

In fact, the main targets for the high-added value products from micro-algae cells are:

- The high purity cosmetics and
- The high quality food supplements

Both of them, concern a new modern market, where the consortium of the VALUEMAG project has the vision to be the market leader!

In this future Initiative, the involved Universities and Research Centers will not be absent: on the contrary, apart from the IPR royalties, subject to a mutual agreement between partners of the VALUEMAG project, the VALUABLE Academic Partners will contribute acting as:

- The technology transfer entities to the consortium of all SMEs
- The characterization and certification laboratories for the above mentioned products and finally The R&D centers of the new consortium-to-be-developed

The project will contribute to the maximization of excellence through the innovation capacity and the integration of new knowledge in several areas of technology as presented in the following Table 2.1:

Technology Area	Ways in which the VALUEMAG contributes	Involved Partners
Development of magnetic nanoparticles	New methods for SPANs synthesis such as microwave assisted flame spray pyrolysis	NTUA & FU SAV
Uptake of magnetic nanoparticles by micro-algal cell and their consequent magnetic activation (MAGMA).	Setting of new protocols based on electroporation to transform a large verity of microalgae and optimization of growth with limited amount of water	EPARELLA – NTUA – NOMASICO – FU SAV
Soft magnetic cone (SOMAC) with the ability of trapping magnetically activated micro algae (MAGMA)	Design, development, characterization and operation of the new magnetic surface	NTUA-NOMASICO-FU SAV
Production of valuable products from micro-algae cells, such as cosmetics, nutraceuticals, food etc	Optimization of super critical fluid extraction & selective magnetic separation	SUN-ENEA-NTUA-FU SAV-NOMASICO-IRIS-THERACELL-EXRGY
Sustainable water re-cycling	Optimization of water use in order to reduce its environmental and economic impacts on algae production	SUN-ENEA-NTUA-FU SAV-NOMASICO-IRIS-EXERGYVERTECH
Sustainable CO_2 capture	Optimization of CO_2 use in order to reduce its environmental and economic impacts on algae production	SUN-ENEA-NTUA-FU SAV-NOMASICO-IRIS-EXERGYVERTECH

Table 2.1: VALUEMAG Innovation Capacity

Competitive advantage of bio-based industries: The value of the global marine biotechnology market in 2011 was €2.4bn with an expected yearly growth of 10%. The consortium will find a huge demand for the VALUEMAG technology after the end of the project. Successful completion of VALUEMAG will provide significant benefits for the partners and for the global community, particularly in the field of cosmetics and food supplements, the new niche and highly appreciated pharma market. This will enable the participating SMEs in the form of the above mentioned consortium, to enter into this market with tremendous growth potential since the VALUEMAG will have an important competitive advantage, covered by Intellectual Property Rights, over current production methods, by providing cost effective algae-based biomass and valuable products, thus increasing revenue in a competitive market (see Section 2.2.7). The VALUEMAG technology will provide products of increased reliability and quality, offering important benefits to several industries (food supplement & pharmaceutical, as well as cosmetics industries). Applications of the proposed innovative approaches can be further used by other sectors (e.g energy, protection of environment etc.)

SMEs will increase/extend/internationalize business activities: The VALUEMAG project technology will enable the mentioned consortium of SMEs to diversify their product/service ranges. The demanding markets have a huge untapped potential worldwide, with an ever increasing number of customers across the globe. As a result of participation in the VALUEMAG project, the participating SMEs will be able to supply the project's results to the world markets. Joining of the knowledge economy through ownership and exploitation of IPR may open new investment avenues including potential new products and processes.

Internal managerial benefits: General benefits include better chances of making correct decisions, improved interagency coordination, and better use of technology, specific benefits, such as justification of programs, would accrue primarily to elected representatives and senior management.

Increasing Human Resources and Qualification: Besides direct economic benefit, VALUEMAG will drive participants to enhance their market position and increase their size, which will mean to mobilize more human resources creating new jobs, mostly high qualified. Within the framework of the project, a series of highly qualified scientists and engineers will have the opportunity for development actions, training, and mobility. By meeting project objectives, the consortium is expected to implement the innovation activities with contribution to workshops, seminars, conferences etc. aimed at sharing knowledge, acquiring new skills and developing careers for consortium staff members in the cross-cutting subject of algae-based bio-refineries. Consortium members, being generally highly internationalized themselves, have long experience and know how to implement new skills development in order to be dynamic player in this promising field. In other words, the involved researchers will become substantially better and most helping in the new consortium, following the successful end of the VALUEMAG project.

Contribution to environmental and socially important impacts

Land use studies suggest that algae make more efficient use of land than traditional crops[45], and one of their advantages over these crops is that they do not require arable land. However, campaigners including Friends of the Earth are concerned that high-volume production of algal biomass could put pressure on arable land, and warn that even land defined as non-arable or degraded may have social or economic importance to local communities. In VALUEMAG technology, limited amount of land/marine/lake area are required because the magnetic photobioreactor chambers are designed to be developed in vertical contrary to the actual open pond systems which extend horizontally. VALUEMAG project will take the advantages of all these evidence and through the proposed holistic LCA analysis will ensure the environmental sustainability and a low environmental impact.

Furthermore, GHG emission from proposed plant will be avoided due to CO_2 recovery. Moreover micro-algal cultivation is an effective and environmental friendly way for a potential extensively sequestration of CO_2, in which 1.8 kg of CO_2 are fixed by producing 1 kg of micro-algae biomass . Promising results to bio-mitigation of CO_2 emission, with CO_2 consumption rate in the range 200–1300 mg/L/day, were showed by several micro-algae strains, such as Chlorella sp., Scenedesmus sp., and Botryococcus braunii . This way, VALUEMAG project offers the

[45] Clarens et al, 2011, Environmental Science and Technology, 44, 1813-1819

possibility to have the best recovery in terms of CO_2 as well as water reuse and could represent the best available technology for CO_2 sequestration from environment as presented in details in the relevant section above.

Social benefits will arise from the development of new and innovative biomass production plants by supporting new job creation and outside investments in local EU communities.

Describe any barriers/obstacles, and any framework conditions (such as regulation, standards, public acceptance, workforce considerations, financing of follow-up steps, cooperation of other links in the value chain), that may determine whether and to what extent the expected impacts will be achieved

Consistent with their diverse characteristics, the development and deployment of algae biomass biorefinery face widely two main market barriers.

- **Economic barriers** include high development and capital costs; limits on access to financing, aside from cost considerations; risks that are difficult to mitigate in regular financial markets, leading, inter alia, to the "valley of death" in commercialization; lack of internalization of environmental externalities of competing, high-emission energy sources; policy barriers that artificially reduce the competitiveness of new technologies; and various types of inertia associated with incumbent technologies. However, the results of the VALUEMAG project will provide all necessary evidence to positively support the decision of the future funds of the consortium-to-be developed.
- **Institutional barriers** include weak institutional capacity to support adoption of new technologies and to monitor and enforce performance standards as well as environmental regulations, information shortages, and cultural and social barriers to doing things differently. The particular and well specified role of each one of the partners, as well as the step-by-step proof of concept will eliminate such possibilities, by allowing all participants to believe in the new under-development methods
- **Regulation and standards issues**: regulation and standards may appear as the weapon of the competitive consortia, following other technologies on micro-algae cultivation and extraction of valuable products out of them. A certain success practice is to engage regulation bodies, such as Classification Societies, as well as Standardization Bodies, such as the International Standard Organization, to discuss and provide regulation and standards on micro-algae production, based on our technology.
- **Public acceptance issues**: the change of the look of sea sides and river sides is certainly an issue in the local people, living in the vicinity of the areas subject to change: despite the straightforward benefits of the new activities to the involved entities and to the Society in general, the local people may see the new algae plant as an obstacle in their every-day life. The solution to that is the involved of the local society in the development, operation, dissemination and exploitation of the new method for algae cultivation.
- **Workforce considerations**: the workforce for the implementation of the VALUEMAG project to a productive mechanism, is an issue, when the local society or the Unions are involved. The solution to this may be the automation process of all production procedures. Thus the recruitment of the workforce can be pretty silent and discrete in order to avoid any unnecessary fights with opposition, sometimes rising from the market competition.

The fact that all these possible weaknesses are foreseen by now, helps in their deep understanding and permits methods and ways to overcome them, as mentioned in the description of each one of them. These corrective actions are completely necessary for the industrialization of the basic ideas and concepts of the VALUEMAG project.

Bio-based production should follow the rules of sustainability. Decisions and policies that would lead to very intensive production in a short-term time horizon should be avoided. An EU-level or even international standardisation and certification scheme, following the above mentioned statement, is essential for the successful development of natural products as part of a bio-refinery mix.

Differential application of schemes can lead to unfair competitive problems and the withdrawal of investment. Effluent from algal biomass production must meet water quality standards, and producers require permits under conditions set out in the Environmental Regulations.

Possible adverse environmental impacts include algal blooms or contamination by heavy metals. These may be accumulated by algae if they are present in seawater, waste water or flue gas. The conditions under which the algae are grown could also affect the regulation of chemicals produced.

The VALUEMAG consortium has identified that the above issues might influence the expected impact of the project results. On the other hand the project itself will clearly contribute to the overcoming of the above mentioned barriers.

2.2. Measures to maximize impact

The approach of VALUEMAG for achieving a sustained impact is presented in the following Figure 2.1.

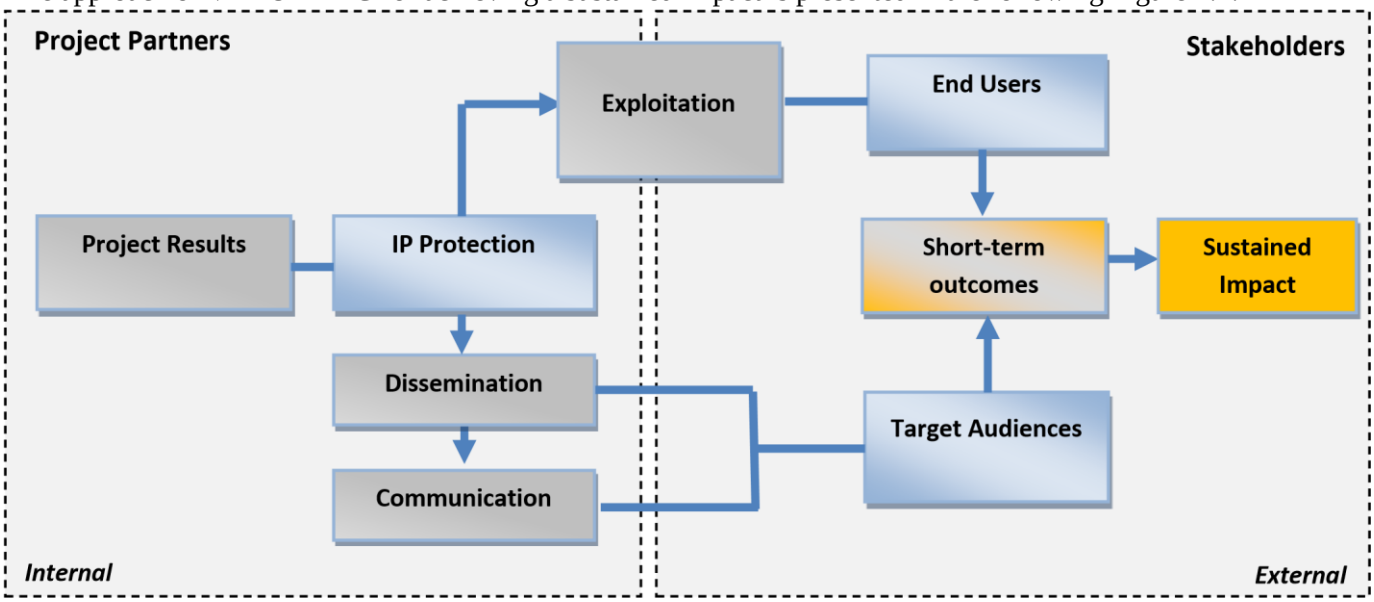


Figure 2.1: VALUMAG approach for sustained impact

2.2.1. Communication, Dissemination, Exploitation Objectives of the Project

The dissemination actions in VALUEMAG are envisaged to communicate and disseminate the activities carried out during the entire duration of the project, the project's main technological achievements and the initiatives organized by the partners of the project within the framework of VALUEMAG as well as their participation in major European and worldwide events and scientific conferences.

The Exploitation actions aim to strengthen and speed up the market uptake of successful results of the project by the development of an exploitation strategy for all eligible results and supporting the partners involved for further exploitation during the different stages of the project.

The following subsections include the objectives solely tied to either the communication or the dissemination objectives and the activities which will be executed during the duration of this project. This distinction is made because the communication objectives and resulting expected actions are more interactive as there are immediate expected responses or other forms of feedback involved. In turn, the dissemination objectives and resulting expected actions are aimed at the spreading of information without the reception of any feedback or responses (such as the distribution of flyers at an event). Some objectives and activities may relate to either category of dissemination or communication but the objectives and activities will not be repeated for the sake of brevity. Instead, the objectives and corresponding activities will be listed under the category in which they relate to most. However, additional measures concerning the industrialization process and the actual demo activities are given in the next sections.

VALUEMAG aims to achieve, within its project duration the following communication, dissemination and exploitation objectives:

Communication Objectives
- To manage the communication network of VALUEMAG. - To take actions to communicate the activities and results of the project - To participate in symposia, meetings as well as organise a Workshop at the end of the project - To create a dedicated website for VALUEMAG project, with public and Members' Only sections - To create all needed communication tools to give a visual identity to the project - To ensure the proper communication and dissemination of the information generated by the project to relevant stakeholders and the general public - To facilitate the communication systems with relevant organizations in order to promote the sharing of data and knowledge.
Dissemination Objectives
- To operate the pilot plant, after getting the patent valid, in a way to act as the optimum advertisement tool of the VALUEMAG project and the under-development industrial consortium made of the involved SMEs.
- To disseminate results of new pathways in the investigation of innovative technologies that will impact on the development of large-scale algae biomass integrated bio-refineries. - To take actions to disseminate the activities and results of the project including high impact publications and road shows and targeting information of important future allies - To increase project awareness via the use of informational and promotional materials (brochures, leaflets etc.) in workshops and conferences in which VALUEMAG will participate. - To disseminate project materials to the widest audience of the general public, to raise awareness around the project topic and create an interest among the audience.
Exploitation Objectives
- Assessment of the expected impact of technology developed, providing a detailed market analysis - Identify eligible results and analyse exploitation potential, by identifying the future key-alliances - Set up exploitation plans for the integrated cultivation and extraction system, as well as individual technologies - Support activities for IPR protection, by enhancing the patents as outcomes of the VALUEMAG project - Decide and prepare the proper business plan to follow the successful termination of the VALUEMAG project

2.2.2. Communication and Dissemination Strategy and Activities

The VALUEMAG project seeks to adopt high impact communication and dissemination activities through efficient and effective communication methods in order to reach the goals/objectives of the project. All of the previously mentioned objectives will be reached by carrying out the listed activities which will be executed via the communication strategies. The main activities, which will be carried out throughout the length of the whole duration of the project include, but are not limited to the following:

Communication and Dissemination tools:

- Preparation of the Communication and Dissemination plan
- Creation and maintenance of project Facebook, Twitter, and LinkedIn accounts
- Creation and usage of videos, which will be posted on the project website and utilized to reach the public at large
- Development of communication tools (logo, posters and leaflets) to support presentations
- Preparation and presentation of the Deliverables
- Distribution of press releases regarding any important objectives and phases of the project as they are reached to be distributed to all media contacts.
- Distribution of an electronic newsletter to target groups and members of the media. **Communication and Dissemination targets:**

- Submission of selected summarizing scientific papers for publication in multidisciplinary high-impact journals. At least one such publication will be prepared in such journal, being one of the best ways of promoting the results of the VALUEMAG project.
- Submission of detailed papers describing the various different processes of the cultivation and extraction process in peer-reviewed journals. Such analysis, together with the above mentioned high-impact publication will attract the interest of the rest of stakeholders in micro-algae cultivation.
- Organization or participation in events, workshops and international fairs in which communication tools will be distributed and project results will be presented to attending researchers and organizations. The consortium considers the initiation of the organization of a regular world congress in the field.
- Organization of one targeted workshop on large-scale algae biomass integrated bio-refineries addressed to stakeholders as well as policy and decision makers in the final months of the project. This event, estimated to take place at the end of the VALUEMAG project will enhance and lighten all project outcomes.
- Organization of training activities, including online training tools as part of the project dissemination plan to spread the generated knowledge.

Communication and Dissemination strategies:

There are two different levels of communication that will be used during different project phases and that determine the creation and adoption of two different types of communication and dissemination tools.

Direct Communication

Communication material will be generated throughout the project lifespan. The tools will be distributed electronically to all the partners, the WP leaders, the BBI and will be also available on the project website, as well as being promoted on the social media pages of the project. The direct foreseen actions are:

(a) Direct mailing (newsletter, press release, project information)

(b) Direct distribution to the stakeholders, included in the Project Contact Repository list, of the communication material produced (leaflets, brochures, printed versions of the newsletters, other information material prepared, invitations to project events and activities, etc.)

(c) Discussions generated on the social media tools

The stakeholders will be reached during the project partners' participation in European and worldwide events organized in the algae relevant sectors, as well as in events related to environment and resource efficiency in general, during the project development.

This direct action is mainly addressed to communicate the project results and to enhance a debate and a discussion on the project with the stakeholders reached. It implies an activity of one-to-one dialogue to give more details on the project and actively involve the target groups in the project.

Indirect Communication

Indirect methods of communication would be used and monitored to ensure consistent communication about the project activities, plans and other information. This activity is carried out from the background point of view that is the generation of the non-confidential content. The indirect communication methods encompass the access by the stakeholders to the following tools, to be realized in a way of stimulating the reader and attracting its attention:

- VALUEMAG project website is the primary medium of communication about the activities and results of the project
- VALUEMAG electronic newsletter, leaflets and other information material circulated VALUEMAG in LinkedIn

Project Communication and Dissemination Tools:

Several tools have been identified and will be developed to facilitate the communication and dissemination actions for the activities. New tools may be added throughout the course of the project according to partners' proposals and will be decided based on a consortium basis. The addition of new tools may also depend on the kind of events the project partners will be attending and the typology of stakeholders to be met. Suggestions will also be taken over

what the best tools are to be used to encourage synergies and co-operation with other platforms and networks or similar initiatives and projects. Thus far however, the main tools which will be utilised throughout this campaign include the following:

- Project Logo
- Project Website
- Project Leaflet
- Promotion sheets / invitations / flyers for project activities
- Project Electronic Newsletter
- Press Releases
- Project PowerPoint template
- Social Media and other online tools
- Project Videos

Target Groups

The target groups involved in the communication and dissemination actions will be part of the following categories:

Non-Scientific community	- Industrial groups related to algae cultivation - Investors and Angel Funds - Policy makers - Associations related to algae - Associations of consumers - SMEs (from all sectors related to algae cultivation) - Public bodies (National, regional, and local) - Civil Society Organizations - High audience Media
Scientific community	- Associations and networks of algae-based industry - Private/public research institutes and universities working on algae - ETPs, agribusiness & marine technology centres, eco-innovation platforms - National and Regional Energy & Enviromental Authorities - National and Regional Food and Agriculture scientific Authorities

Each of these categories will be addressed according to the type of dissemination material to be distributed and the type of information prepared to be circulated.

A mailing list with all contacts, Project Contact Repository, gathered among all the participants has to be prepared in the initial phase of the project and used with respect to the objectives of the VALUEMAG project.

The Project Contact Repository of all the targets will be setup with the contribution of all partners. It will include all available contacts details with a minimum of the e-mail addresses. Each partner will provide contacts related to its area of expertise, related sectors and/or geographical area. The list, to be regularly updated and checked, will be at disposal of all and only the partners of the consortium in the members' area of the website. New contacts can as well be inserted through an online self-subscription on the public area of the VALUEMAG website. For gathering contacts, the partners will use their own connections and channels and also international networks.

2.2.3. IPR management and Exploitation Strategy and Activities

Throughout the project, all partners will continuously contribute to the identification of project results that may qualify for IPR protection. In the VALUEMAG project new IP and subsequent patenting is anticipated in the following areas of technology:

- Methods of developing better, faster and cheaper magnetic nanoparticles and related surfactant covering films and ligands
- New compositions of magnetic nanoparticles and related surfactants & ligands
- New techniques for the uptake of magnetic nanoparticles by micro-algal cell and their consequent magnetic activation (MAGMA)
- New methods, devices and instruments for the optimum operation of the soft magnetic cone (SOMAC) with the ability of trapping magnetically activated micro algae (MAGMA)
- New methods to produce valuable products from micro-algae cells, such as cosmetics, nutraceuticals, food etc.
- Sustainable water re-cycling Sustainable CO_2 capture

In order to avoid problems related to IPR issues within the consortium, special attention will be paid to the specific IP paragraphs in the Consortium Agreement to be signed. However, all IPR issues will be treated under the view of the future consortium of transforming the findings of the VALUEMAG project to an industrial production line.

Basically, the general principles for Intellectual Property Aspects, set out by the EC for HORIZON2020 projects apply. Existing know-how (background or pre-existing intellectual property) of a specific partner shall be made available on transfer conditions to the Partner (or Partners) within the consortium that need this information for the proper execution of their tasks within the scope of the project. The use of such existing know-how is strictly limited for use to the achievement of the project goals and for the duration of the project. The receiving partner or partners will sign appropriate confidentiality agreements from the providing partner.

An overview of the existing know-how will be included as an annex to the consortium agreement. All partners shall be entitled to license existing know-how to the extent necessary to exploit their own results. Licensing of existing know-how to third parties will be done on commercial terms and conditions whereas licensing of existing knowhow to partners of the consortium will be done on fair and reasonable terms and conditions, under the view of the future development of the Industrial Consortium of the existing partners.

Foreground (e.g. results, including intellectual property generated during the project) shall be owned by the partner or partners who developed the results. Each partner is responsible for taking the appropriate steps for securing intellectual property of the knowledge or results created during the project (e.g. filing of patent applications). Each partner is obliged to fully inform the exploitation manager and the project coordinator of the filing of patent applications of knowledge or results created in the field of the project within two weeks of the data of filing, in order to strengthen the future collaboration of partners towards the industrialization of the VALUEMAG results. Although any owner of any result shall be free to use such result and to commercialize and distribute it, the consortium understands that the power of the industrialization of the VALUEMAG findings is exactly in the assembly of all SMEs of the project in one future consortium, aided by the four involved academic labs (two Universities and two Research Centers), towards the optimum exploitation of the VALUEMAG findings. However, in case of joint ownership, each owner shall be free to use such result, provided all its joint owners agree and the related action is according to all signed agreements.

Project results shall be made available free of charge to partners of the consortium for research purposes within the scope and the duration of the project. VALUEMAG results owned by one or more of the partners shall be licensed to other partners of the consortium on favorable conditions to the extent necessary to enable these partners to exploit their own results and mainly to aid the future collaboration on the industrialization of those findings. The exploitation strategy will be based on:

- Appropriate management of the knowledge generated in the project and IPR protection according to the interest of the partners,
- Assessment of the expected impact of each technology developed and the factors that could influence its exploitation (standardization, required level of knowledge, regulatory aspects, etc.),
- Analysis of the exploitation potential of the technologies, reviewing all aspects from the viewpoint of potential investors, internal or external, and marketing people.
- This will allow to check carefully the strengths and weaknesses of the results as future business consortium, and to identify all other necessary resources and actions required to exploit them. Incorporation of individual exploitation plans for all relevant partners.

In order to bootstrap on top of the work conducted during the VALUEMAG project and ensure a timely entry and expand into this rapidly developing market it will be necessary to find further funding for VALUEMAG. This task will be organized by dissemination and exploitation council. There are four main potential sources of funding:

- EU funding for further innovation projects
- EU or national funds for business development and technology exploitation
- Venture capital or investment funds
- Public Private Partnership initiatives
- Particular engagement with significant, existing stakeholders in micro-algae cultivation

Funding may come from a mixture of all five sources, but it should be noted that further funding focused solely on innovation would be unlikely to be enough on its own. Whichever funding sources VALUEMAG is able to access, it goes without saying that the concrete, detailed business plan will be necessary. Under this prism a business plan will be prepared during the project with some practical aspects on the market strategy. According to the interaction with internal and external environment during these sessions, the business plan delivered at the end of the project will contain all necessary details on the industrialization and commercialization of the proposed technology.

2.2.4. Plan or the publications and presentations

It is expected that several scientific papers will be published during the implementation period of VALUEMAG in the following fields:

- Transformation of microalgal cell with magnetic nanoparticles
- Cultivation of magnetically immobilized microalgae
- Optimum development of the SOMAC
- Operation and productivity analysis of the SOMAC
- Production of cosmetics, food and nutraceuticals from algae cells in cost effective ways New methods of producing cosmetics, food and nutraceuticals from algae cells Water re-cycling and CO_2 capture in a cost effective and sustainable way.

The VALUEMAG consortium commits to release tailored publications showcasing project outputs, in **peerreviewed journals** including **open access peer-reviewed journals** or specialised magazines. They will be prepared each time the project has key findings to disseminate. In addition, project partners will possibly contribute to e-Journals, blogs and newsletters targeting a larger public with shorter articles and news, as well as to policy oriented publications to enhance project outreach to policy-makers. However, a major target is the preparation and publication of a high-quality breakthrough paper in a multi-disciplinary Journal of as high as possible impact factor, indirectly illustrating the necessity to industrialize and commercialize the findings of the VALUEMAG project, such as “Nature” or “Science” or the like. The plan for these activities is the following:

- Over the project duration: project partners commit to release at least 10 publications, demonstrating the VALUEMAG scientific findings;
- Over the project duration: at least one publication in a multi-disciplinary high impact Journal, summarizing all scientific findings and stretching the necessity of industrialization and commercialization
- Each time they identify one interesting Call for papers/articles they wish to apply to, the partners are invited to complete the internal online table to provide information on;
- Before submitting a scientific publication, partners are invited to send a draft version to the consortium members According to Art. 29 of the Annotated Model Grant Agreement of the European Commission. “Beneficiary that intends to disseminate its results must give advance notice to the other beneficiaries of - unless agreed otherwise — at least 45 days, together with sufficient information on the results it will disseminate”;
- Therefore, “Any other beneficiary may object within — unless agreed otherwise — 30 days of receiving notification, if it can show that its legitimate interests in relation to the results or background would be significantly harmed. In such cases, the dissemination may not take place unless appropriate steps are taken to safeguard these legitimate interests »;

- According to Art. 29 of the Annotated Model Grant Agreement of the European Commission. « Each beneficiary must ensure open access (free of charge, online access for any user) to all peer-reviewed scientific publications relating to its results”; In particular, partners must:

  a) as soon as possible and at the latest on publication, deposit a machine-readable electronic copy of the published version or final peer-reviewed manuscript accepted for publication in a repository for scientific publications; Moreover, the beneficiary must aim to deposit at the same time the research data needed to validate the results presented in the deposited scientific publications.
  b) ensure open access to the deposited publication — via the repository — at the latest:
  c) on publication, if an electronic version is available for free via the publisher, or
  d) within six months of publication in any other case.
  e) ensure open access — via the repository — to the bibliographic metadata that identify the deposited publication.

  The bibliographic metadata must be in a standard format and must include all of the following:

  - the terms “European Union (EU)” and “Horizon 2020”;
  - the name of the action, acronym and grant number;
  - the publication date, and length
- Each publication will be designed into an eBook, that will be disseminated on internal and external channels;
- All partners contribute to the promotion and dissemination of the various publications.

Partners will make efforts to be invited as speakers and to give presentations in already organized conferences and events. The presentations will focus on presenting project’s outcomes and expected results.

Partners will inform the Dissemination and Exploitation Council of any upcoming events they may be contacted about in which VALUEMAG may participate in or present in a timely manner.

Upcoming events will be posted on the project website and will be updated regularly as new events are announced. The partners suitable for this task will be selected and destined to conferences related to the specific competencies and activities the same partner is in charge of. The selected partners will prepare the presentation for speeches that will be monitored by the Dissemination and Exploitation Manager.

The list of conferences the partners will attend will be explored at the beginning of the project and updated at each project meeting.

2.2.5. Open Workshop

The main findings of the project will be presented at an Open Workshop in Brussels. This workshop will be the occasion to present the project results and disseminate information to all stakeholders. The participants will include SMEs, industries, organizations involved in the algae production chain, consumers associations, researchers, policy makers and investors. The participation expectation is up to 300 participants. It will also imply a moderated Round Table discussion on the exploitation of project results in the format of a round-table discussion to prepare the final users on the implementation of the project results. The involvement of media will be of particular attention to increase the opportunities of visibility among the end-users, suppliers, groups of consumers and the specialists in the area

2.2.6. Building on the results from previous and on-going research projects

Worldwide, research and demonstration programmes are being carried out to develop the technology needed to expand algae biomass production and exploitation. VALUEMAG consortium will exploit the available knowledge and will create a network of knowledge with the responsible bodies of these projects. In the following, we mention some of the most relevant projects concerning the VALUEMAG project:

Project Title	Area
Biofuel from Algae Technologies	BIOFAT
Sustainable Fuels from Marine Biomass Project -	BIOMARA

Bio-waste and Algae Knowledge for the Production of 2nd Generation Biofuels	BIOWALK4BIOFUELS
Energetic Algae	ENALGAE
Utilisation of Micro-algae for Wastewater Treatment with Energy Purposes-	ENERBIOALGAE
Marine Algae as Biomass for Biofuels	MABFUEL
Biotechnological Exploitation of Marine Products and By Products	BIOTECMAR
Control of Light Use Efficiency in Plants and Algae From Light to Harvest	HARVEST
Towards a Better Sunlight to Biomass Conversion Efficiency in Micro-algae	SUNBIOPATH
Enabling European SMEs to Remediate Wastes, Reduce GHG Emissions and Produce Biofuels via Micro-algae Cultivation	BIOALGAESORB
Project Title	**Area**
Development of an algal platform for production of building blocks for the chemical industry and their further conversion to products	SPLASH
A sustainable chain for continuous biofuel production using micro- algae as a production platform, thereby making 2nd generation biofuels competitive alternatives to fossil fuels.	Fuel4Me
Demonstration of integrated and sustainable enclosed raceway and photobioreactor microalgae cultivation with biodiesel production and validation.	INTESUSAL
Pilot scale algal research centre (including lipid production) Relevant to Biology & Biotechnology, Economic assessment	ALGAEPARC
Novel algae-based solution for CO_2 capture and biomass production	ALGADISK
Genetic Improvement of Algae for Value Added Products	GIAVAP
Sustainable Production of Biologically Active Molecules of Marine Based Origin	BAMMBO
Oil production with algae	Emerald Oils

2.2.7. An Initial Market Analysis

Bio-Based Industry is already a sector of 3,2 million jobs. According to NOVA institute, the global bio-economy is worth € 2.1 trillion, with the EU28 bio-based market already at € 600 billion today. The bio-based industries have ambitious but realistic targets, aiming to replace 30% of the oil based chemicals and materials in Europe by 2030. The bio-based market can increase diversification and growth of farmers' income by adding up to 40% additional margins. It's estimated that around 700,000 new jobs will be created between 2013 and 2030 by bio-based industries in Europe. A high number, around 80% of these new jobs, will be in rural areas – good for farmers, their local economy, and for regional regeneration and redevelopment[46].

Within the maritime sector algae production hold great promise. Although the total production volume and market size of micro-algae in general are still relatively small, they have been characterised by high growth since 1999. In 1999, global production volumes of micro-algae were estimated at only 1000 tonnes dry weight. This has increased

[46] Nova-Institute for Ecology and Innovation, European Bioeconomy in Figures, March 2016. http://biconsortium.eu/sites/biconsortium.eu/files/news-image/16-03-02-Bioeconomy-in-figures.pdf

to 5000 tonnes dry weight representing 1,2 billion euros by 2004[47], which represents a 5-fold increase in five years. In 2011, the total production volume has risen to 9000 tons dry weight[48].

Furthermore, collaboration with food additive and neutraceuticals can provide the cash flow needed to sustain the sector until demonstration plants develop into commercial scale biorefineries for algae production. The value of the global marine biotechnology market in 2011, with micro-algae as its main component, was estimated at €2.4bn, with an expected yearly growth of 10% [49]. Note that this volume is still small compared to other food commodities. (global wheat production for instance is around 700 million tons annually -70.000 times as much). Over the past decades, over 75% of the production volume of micro-algae was used in the health food market as dietary supplements[50]. Large companies in the food ingredients market - BASF, Unilever and Dow Chemical - have noticed the potential of these applications and their associated growth and have made major acquisitions.

Table 2.2. shows the most recent information available on the different sub-markets of micro-algae(-derived) products. Key players are mentioned between brackets in cases where the total production volume is dominated by these specific producers. The potential market for derived product is measured by the market size of synthetic (e.g. Astaxanthin) or traditional (e.g. fish oil) alternatives. The results clearly show that Spirulina and Chlorella still have by far the largest production volumes, but also point to the large potential markets for high-value products such as DHA/EPA, β-carotene and Astaxanthin. In general, the production is quite concentrated on a small number of players, except for Chlorella production that is distributed across relatively many small players. All data are world estimates, and due to the opacity of production volumes of individual producers, shares for specific countries/regions cannot be deducted. Information on the size of the European market is therefore not available.

Current product based on micro- algae	Production volume (tons/ year dry weight)	Number of producers (key players)	Value of production volume (yearly turnover)	European share in production (%)	Potential market (synthetic / traditional forms)
Food and feed products: whole dryed micro-algae biomass					
Spirulina	5,000 tons/year (2012)	>15 companies (Cyanotech / Earthrise)	US $ 40m (2005)	(NA)	No synthetic alternative
Chlorella	2,000 tons/ year (2003)	> 70 companies (NA)	US $38m (2006)	(NA)	No synthetic alternative
Food and feed products: micro-algae components					
Astaxanthin (based on Haema¬tococcus)	300 tons/year (2004)	>8 companies (Fuji Chemicals, Cyanotech)	US $10m (2004)	(NA)	US$200m (2004)
Phycibiliprotein colourants (incl phycocyanin)	(NA)	>2 companies	(NA)	(NA)	> US $ 50m (2004)
EPA/DHA (Omega-3 PFA) (based on Chrypthecodinium)	240 tons/year (Martek) (2003)	>4 companies (Martek/DSM)	> US $300m for Martek (2004)	(NA)	±US 14.39bn (2009)

[47] Milledge, J. Microalgae - commercial potential for fuel, food and feed. 2012. FS&T Vol26, Issue 1 (2012). http://eprints.soton.ac.uk/336243/

[48] Acién Fernández, F G; Fernández Sevilla, J M; Molina Grima, E. Reviews in Environmental Science and Biotechnology12.2 (Jun 2013): 131-151.

[49] Guedes et al. Microalgae As Sources of High Added- Value Compounds—A Brief Review of Recent Work. 2011. Biotechnology Progress Volume 27, Issue 3, pages 597–613, May/June 2011

[50] Chacon-Lee, T.L. & Gonzalez-Marino, G.E. Microalgae for Healthy Foods - Possibilities and Challenges. 2010. Comprehensive Reviews in Food Science and Food Safety. Volume 9, Issue 6, pages 655–675, November 2010.

β-Carotene (based on Dunaliella Salina, Schizochrytium, Nannochloropsis)	1,200 tons per year (2010)	> 10 companies (Cognis/BASF)	(NA)	(NA)	US $ 285m (2012)
Other products					
Stable isotope biochemicals	(NA)	> 1 company (Martek/DSM)	> $13m (2006)	(NA)	(NA)
Skin products	(NA)	> 4 companies			
(Solliance, Solazyme/ Unilever)	(NA)	(NA)	(NA)		
Fluorescent protein markers	(NA)	> 1 company			

Table 2.2: Market figures of micro-algae based food products[51]

Therefore, the VALUEMAG project can target although not exclusively, the following main products out of SOMAC operation that can be the basis of the under development business plan during the VALUEMAG project:

- Dietary supplement
- Aquaculture food
- Lutein
- omega-3
- DHA/EPA
- β-carotene and Astaxanthin

The quantitative analysis of these products, as well as other valuable products from the micro-algae cultivation is a deliverable of the VALUEMAG project. **The SOMAC demo plat in the frame of VALUEMAG project will operate with different species of micro-algae for a limited time (3 months, e.g.) to test their productivity and the extraction yield.**

However, the produced de-watered micro-algae is also a subject of commercial production, provided that proper agreements between the consortium and related stakeholders are achieved.

A most acceptable but not exclusive way of the industrialization and commercialization process of all these products is the B2B action.

In any case, all productivity projections, shall be annexed in the above mentioned business plan, subject to be developed within the VALUEMAG project as a discrete deliverable.

3. Implementation

3.1. Work plan — Work packages, deliverables

We have designed a comprehensive project work plan, to ensure that all of the scientific and technological needs of the VALUEMAG project are addressed and the project deliverables are fulfilled. This work plan has been split up into six (6) work packages, each resulting in key deliverables. Figure 3.1 shows interconnections between these work packages and Figure 3.2 is a Gantt chart showing the project's timeline.

Management and disseminations actions are grouped within WP6 while exploitation activities are considered within WP5. The WP1, WP2, WP3, WP4 deal with interconnected scientific and technological activities in order to demonstrate the system. This structure is designed to maximize the interaction between all participants (more details in $3.3). A short description of each work package follows below:

[51] Enzing, C.; Ploeg, M.; Barbosa, M.J.; Sijtsma, L., Microalgae-based products for food and feed sector: an outlook for Europe, Joint Research Centre, 2014.

WP1- Magnetic algae preparation: The main aim of WP is to set up protocols to transform microalgae cell with magnetic nanoparticles using mainly electroporation. Viability of transformed cell will be check and optimized.

WP2 - Magnetic cultivation of Algae: WP2 will focus on the on the fabrication of SOMAC surface and on the optimization of MAGMA growth onto magnetic surface.

WP3 - Animal Food, Nutraceutical & Cosmetics from Algae: Bio-products derived by micro-algae will be processed in order to produce cosmetics, animal food and nutraceutical by supercritical CO_2 extraction and by a selective magnetic separation based on functionalized magnetic beads. Moreover, scaling-up of CO_2-SF technology for bio-product processing will be carried out.

WP4 –Operation and Assessment: The aim of WP4 is to set the demonstration into operation, evaluation it and perform the environmental impact assessment from its operation.

WP5 – Exploitation and Business Development: The aim of WP5 is to define and execute exploitation activities to maximize the impact of the results of the VALUEMAG Project. In this WP Project Board will explore the opportunities for future financing from private or EC funding sources.

WP6 – Project Management & Dissemination: This work package is responsible for ensuring that the project has an effective project management structure and that project activities meet appropriate quality levels, checking and validating the correct scheduling of tasks and disseminate the outcomes of the research work.

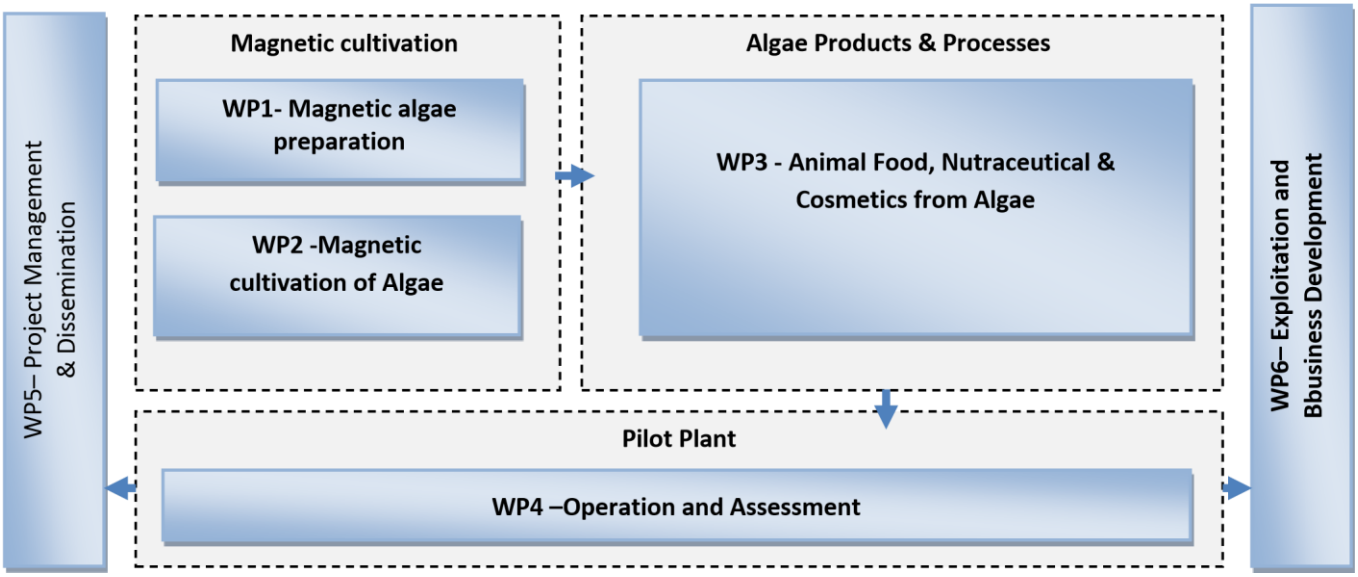
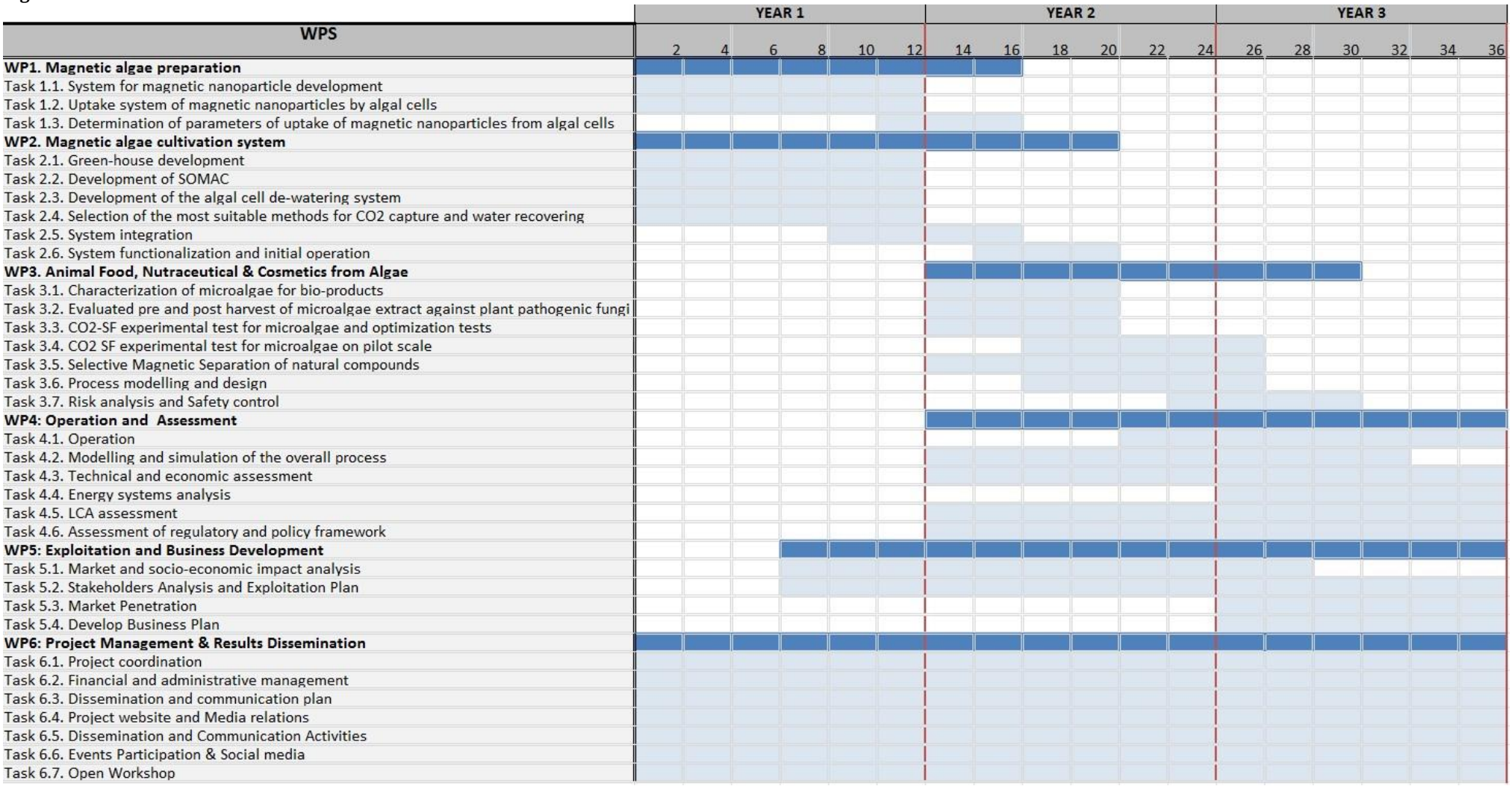


Figure 3.1: Graphical presentation of the Work Packages showing their interdependencies

Figure 3.2: Gantt chart



3.2. Management structure and procedures

3.2.1. Management structure

Management structure is set up to ensure that important decisions are made effectively and without delay, while ensuring that project management is correctly handled and technical goals are achieved without issue. The management structure clearly distinguishes between Technical Management and Administrative Management.

Technical/Scientific Management tasks involve all partners and deals primarily with the coordination of RTD and Demonstration Activities, as well as Other Activities (such as dissemination and exploitation). Technical management doesn't require a specific work package, as day-to-day technical decisions will normally be made by the partners involved in each WP, and each WP will have a clear WP Leader.

Administrative Management (administrative, legal and financial management) has a dedicated work package (WP6) to deal with financial, legal, ethical and administrative tasks. The coordinator NTUA will be the supervisor for these management tasks, ensuring that the rest of the partners meet the requirements on time. Control of resources; communication policy among the partners; dispute resolution; respect of the rules set in the Consortium Agreement and fulfillment of BBI-JU requirements with regard to reporting and communications are included in this WP. To successfully manage an international project of this magnitude, a clear and well defined management and decision making structure is required. We have outlined this structure below and discussed how it has been designed to focus the project on our needs.

Project Board: The Project Board has been empowered to ensure the overall success of the project and resolve any contractual issues within the consortium and for the BBI-JU. The Board is responsible for overseeing the development of the project and reviewing its progress in accordance with each work package, milestone, objective and deadline. The board will follow a gated management system which will enable them to control the direction and speed of the project, by signing off the completion of each work package and the performance of each partner involved.

Project Coordinator: Outside of the board meetings the Project Coordinator will be responsible to administer the contractual, legal, financial, logistical and secretarial aspects of the project. The entire consortium will be responsible to the BBI-JU for producing the project deliverables, supplying them to the BBI-JU according to the timescales described in the contract, and for consolidating the project plans, progress reports, cost statements and budgetary overviews. Professor Evangelos Hristoforou has been appointed to the position of Project Coordinator for the VALUEMAG Project.

Technical Manager: The Technical Manager will support the Project Coordinator by monitoring the progress of each of the work packages and coordinating day to day activities between each of the Work. The Technical Manager is responsible for managing the project risk log and contingency plans and will therefore also be acting as the project Risk Manager also. As a result of this role requiring technical expertise in addition to experience in international project management.

Dissemination and Exploitation Council: The Dissemination and Exploitation Council is made up of representatives from each of the consortium member. The role of the council is to direct the future development and commercial success of the project and recommend actions to the Dissemination and Exploitation Manager.

Dissemination and Exploitation Manager: The role of the Dissemination and Exploitation Manager is to ensure the future and commercial success of the project, whilst managing the IPR under the guidance of the Council. Dissemination material such as technical papers and presentations must be approved and submitted to the relevant journals and technical institutions. In addition they will responsible for data management during the project lifetime. IPR issues will be formalized in the Consortium Agreement, which will deal with all aspects of the ownership and exploitation of IPR.

Work Package Leaders: The main investigative element of the project is organized as a set of closely related packages of work. WP Leaders have been identified for each of these and will be responsible

for co-ordination, planning, monitoring and reporting all work package activities. The WP Leaders will organize independent, technical meetings to discuss the work share between involved partners, technological problems and new results in detail. Each WP is divided into a number of Technical Tasks led by the partner best able to carry out that task. The Work Package Leader will collate the results of the individual tasks and report them to the Project Coordinator, who will circulate these reports to the Project Board and incorporate them into the periodic reports to the BBI-JU, as appropriate.

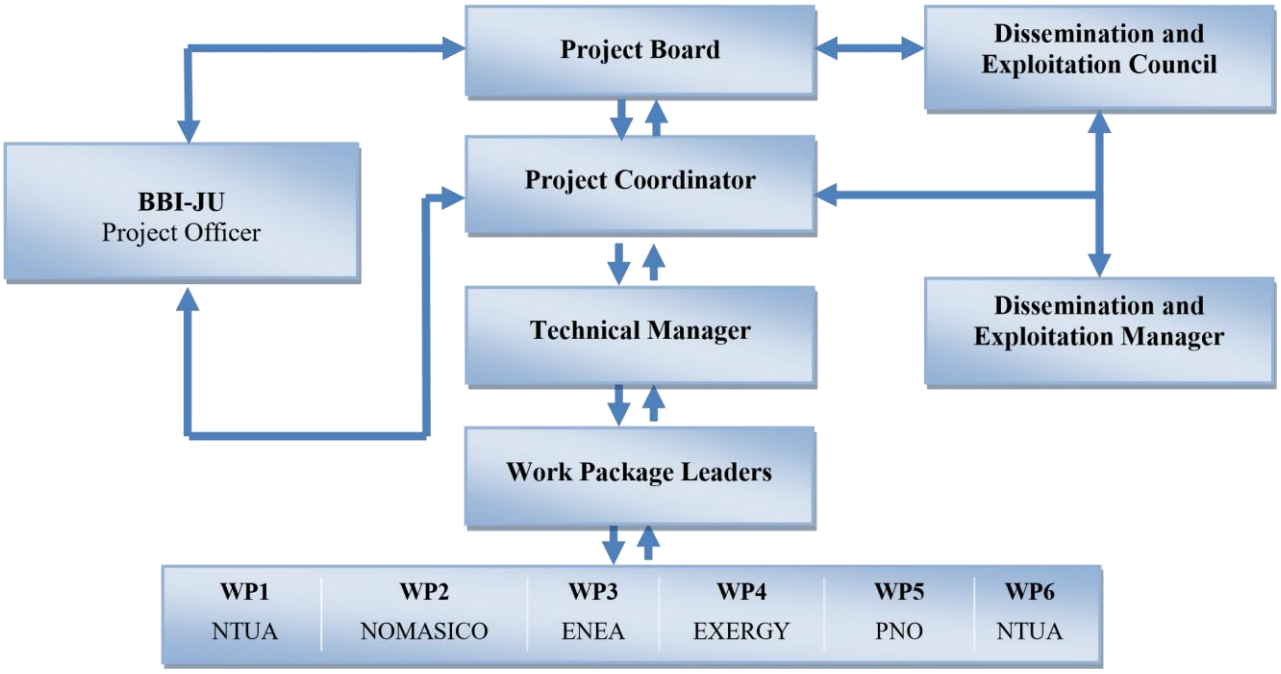


Figure 3.3: Project Management Structure

3.2.2. Management procedures

Monitoring progress and quality: Progress monitoring will be done via reports and meetings, including periodic visits by the Project Coordinator with Work Package Leaders. The Coordinator will frequently update the Project Board on progress. Project results must be correctly reported, and to that end the Coordinator must ensure that all partners keep detailed records of input into the project using an agreed upon system (logbooks, time cards, etc.). An assessment form will be created, and will include the duration, work package identification, resource consumption, deliverables and an assessment of the tasks. This form will be prepared on a monthly basis, and will be sent to the Project Board. The work package leader will be responsible for collecting the information from all partners, making the assessment, and preparing this form. At midterm period, a report will cover the entire project, including a breakdown of the input of each partner.

Furthermore, to ensure project quality, a continuous quality control mechanism will be defined. The role of quality monitoring of project work and deliverables will be assigned to the Technical Manager, whose initial focus will be the definition of suitable project quality standards. This will subsequently be followed by deliverable assurance assessment as required. Specifically, quality assurance activities will be:

- Defining project quality standards and verifying the performed activities against them.
- Defining a Quality Assurance procedure for project deliverables including a number of pre-approval steps and activities required to apply the above quality standards.
- Defining a peer-review procedure of interim and final project results in which the Technical Manager and principal users of project deliverables will be involved, evaluating their quality.
- As part of standard project quality procedures, all deliverables will be made available for review to the experts from the Project Board.

Project Meetings: The project partners convene for regular project meetings to discuss the project, its progress and future, as well as exchange experiences and updates. Each partner is represented at these project meetings. The Coordinator is responsible for organising the meeting, the practical part of this responsibility may be delegated to a partner on a per-meeting basis. More than one person per partner may attend the project meetings. The tentative meeting schedule is that the project meetings and board meetings will be held about semiannually, rotating between partner locations, pending logistical considerations. These meetings will comprise reports on scientific and administrative progress and issues, discussions, the exchange of knowledge and project decisions. When applicable and legally possible, the project will invite experts to attend parts of the meetings to comment, give advice and share knowledge. The coordinator chairs the project meetings. The Coordinator is responsible for distributing the agenda and meeting details at least two weeks before the meeting. The following project meetings are foreseen. The final meeting calendar will be confirmed at the kick off meeting that will be held within the first 30 days of activities at the premises of the coordinator.

Nb	Meeting type	Month	Location
1	Kick-off meeting	M1	Greece
2	Project meeting	M6	Italy
3	Project meeting + stakeholders meeting	M12	Belgium
4	Project meeting + mid-project review	M18	Cyprus
5	Project meeting	M24	United Kingdom
5	Project meeting + stakeholders meeting	M28	Spain
7	Project meeting + final review	M36	Belgium

Decision-making mechanisms and processes: In VALUEMAG three main types of decisions are foreseen:

- Strategic, higher-level decisions, (typically involving issues related to the overall strategy or policy of the project), contractual matters, and other items with an impact on the entire project will be handled during meetings, where each representative has one vote and a general majority is required to pass a motion. The basis for decisions will be prepared in advance by the Project Coordinator or Technical Manager (depending of the WP(s) involved) and normally debated in advance during project meeting.
- Operational matters that span several work packages and have an impact on parts of the project will be agreed through consensus or majority decisions in suitable forums. Issues will be discussed broadly at project meetings, via tele-meetings or e-mail involving relevant partners. Either the Project Coordinator or Technical Manager (depending of the WP(s) involved) will be responsible for facilitating such discussions and ensuring that a decision is made and communicated to those affected by it.
- For matters that pertain only to single tasks or one work package, decisions will be made by work package teams, chaired by the Work Package Leader, or internal teams to a partner. For such decisions WPLs are responsible for facilitating a transparent decision process.

Should situations arise where it is proving difficult to make a decision that all parties involved can accept, conflict resolution activities will be initiated. The first approach is to attempt the resolution of the situation by group or bilateral discussions focusing on isolating the problem and finding constructive ways/compromises to handle it. In extreme conflict cases, where agreement cannot be achieved, the project board will organise a conflict resolution meeting, either on its own initiative or following request by a partner. Such a meeting will be held with all necessary representatives; the quorum requested for arranging a meeting and making decisions will be defined according to the conflict's nature and the number of organisations involved. Within the meeting, resolution will be

sought through dialogue and mutual compromise between the partners. In the case of failure, a meeting at the upper project management level will be arranged. If it becomes necessary to involve the responsible Project Officer, a formal request for a meeting will be submitted. Such a request will include potential solutions and will request an answer within a stated time. During the project's life, disputes or differences arising in connection with the Consortium Agreement or with the distribution of tasks from the Technical Annex, may be resolved by mediation if they cannot be settled with the support of the Project Coordinator.

Knowledge and Innovation Management: A major concern for the VALUEMAG consortium is to identify, which innovations generated by the work packages can be transferred to society, and in particular, to industry. The successfully demonstration of the new algae-biomass based bio-refinery will generate high-volume industry contracts and boost industrial developments in various high-tech domains. Therefore, one of the main goals is to prepare the grounds for a long-term industrialization plan in view of construction and operation phases. These goals require active technology scouting, identifying competitive technologies. Potential followup industrialization and dedicated R&D programs, which are of particular interest for industry in the coming decade will be identified. Performing these tasks requires networking at several levels: Communication with all participants is mandatory throughout the entire duration of the project to identify key areas that can be the subject of technology transfer. Existing networks will be used as tools of information dissemination to all the stakeholders. A particular Workshop is planned to bring together the consortium and interested industrial partners to explore the technologies developed in the framework of the project.

VALUEMAG will ensure 'gold' open access to all peer-reviewed scientific publications relating to its results. Immediately upon acceptance for publication, a machine-readable electronic copy of the manuscript will be deposited into an OpenAIRE sponsored repository. At the same time, participating in the Pilot on Open Research Data in Horizon 2020, VALUEMAG will make the underlying data needed to validate the results presented in scientific publications available for use by other researchers, industries and citizens, by depositing them into a data repository with Creative Commons Licence attached. In addition, the consortium ensures open access, via the repository, to the bibliographic metadata that will identify the deposited publication.

VALUEMAG aims at demonstrating new technologies with potentially high commercial value when exploited. To avoid dissention and conflicts inside the project team, both existing rights to pre-knowledge and ownership rights to items developed during the project must be well managed. So as to handle these issues, the following legal agreements will be implemented:

- A consortium agreement, addressing issues such as intellectual property rights (IPR), confidentiality issues, voting rules, monetary obligations, etc. This will be based on similar previous project agreements, but adapted according to the specific objectives of VALUEMAG.
- A commercialisation agreement to regulate exploitation and commercialisation If necessary, any specific documents concerning confidentiality.
- At the beginning of the project, IPR terminology will be defined in the Consortium Agreement. The state of the background, the rights to use the foreground generated in the project and general access rights will then be the subject of specific IPR clauses which will be developed according to the recommendations of the BBI, following the rules and regulations of the Horizon 2020.

Risk and Contingency Management: VALUEMAG is a technically challenging project and contains an inherent amount of risk. It has therefore been important to design a management procedure to handle these risks and to identify a series of contingency actions in order to minimize their impact. Project participants are required to maintain a risk log, which contains potential risks and a record of the occurred events. This information will be used to identify any issues and future risks which may occur during the implementation of the project. Risk will be managed initially by breaking the project into an achievable set of smaller developments, enabling participants to monitor progress and measure the

progressive reduction in risk. The risk management process that the VALUEMAG consortium will follow is defined in Figure 3.4.

The risk management process will be iterated on a monthly basis to ensure that:

- Identified risks are continuously monitored until judged acceptable, Regular evaluation is performed to identify new sources of risks, Risk mitigation actions are performed and monitored.

All project members, and especially Work Package Leaders, are responsible for identifying new areas of risk to the completion of the project. Any person identifying such a risk will write a risk analysis report and send it to the rest of the partners. The Project Coordinator, in consultation with the Project Board, will prioritise the new risks identified.

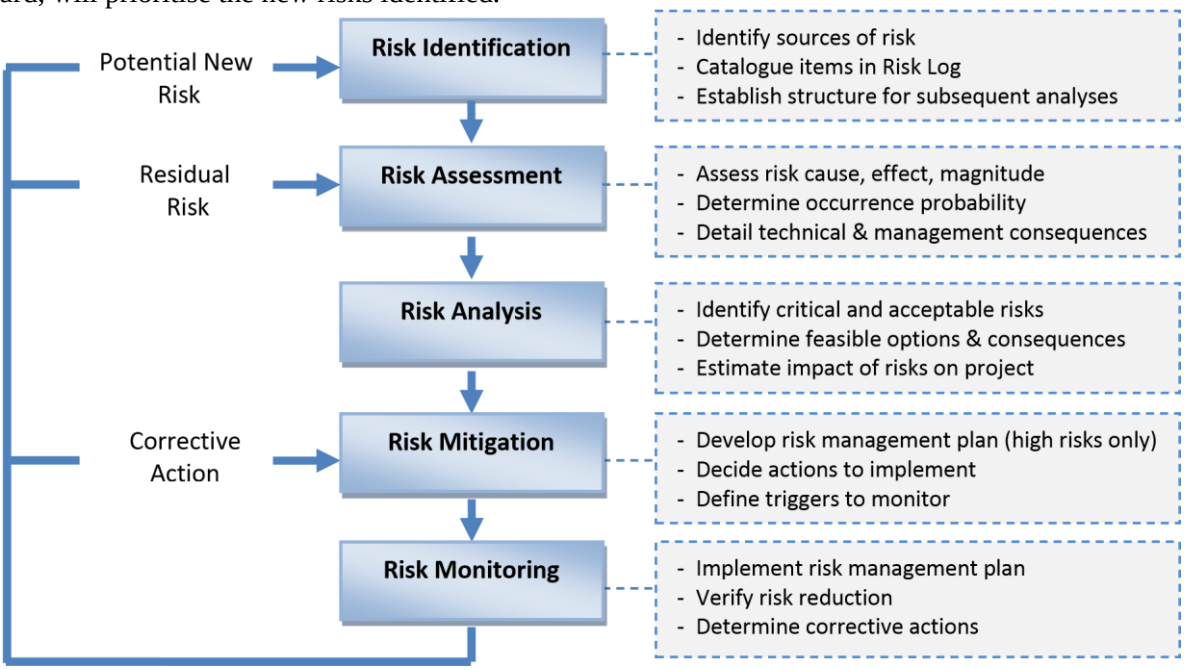


Figure 3.4: Risk Management Process

We have completed an initial risk assessment for the project and we have included a summary of the report in the table below. The probability of a risk occurring has been estimated, along with the individual and combined impacts on the remainder of the project deliverables. It is important to address the potential foreseeable risks within the project and how their contingency actions may be integrated to minimize their impact.

3.3. Consortium as a whole

In order to optimize the value of the VALUEMAG project, all the experts of the value chain for development and end user applications are involved in the project. The VALUEMAG consortium consists of 11 participants from 9 European countries who provide all the necessary background and expertise to achieve the objectives of the project. The VALUEMAG Project combines:

- Leading research organizations with extensive experience in the continuous development of next generation technologies that combine top class academic know-how with a long tradition in providing turnkey solutions to major European industries (SUN, NTUA, ENEA, FU SAV)
- Market-oriented industrial actors with international experience in the implementation of very large projects and the launching of new products and services (PNO, VERTECH) and
- Highly specialized high tech SMEs (NOMASICO, EPARELLA, THERACELL, IRIS, EXERGY)

Specifically,

NTUA: has strong impact on magnetism, magnetic materials and mainly their applications; has vast experience in producing magnetic nanoparticles and using them as the driving for navigation purposes.

NTUA owns the pending patent on the soft magnetic cone (SOMAC) and the corresponding magnetic micro-algae cultivation method. NTUA will produce the required nanoparticles; aid NOMASICO to the electroporation and development of the Cultivation chamber and the pilot plant in Cyprus; aid in the operation of the pilot plant.

SUN: Due to its multidisciplinary, SUN will offer expertise in several scientific fields such as chemistry, environmental science, civil-environmental-chemical engineering. Moreover, thanks to its experience in chemical and environmental processes and strong know-how in thermodynamics, kinetic and mass and heat transfer, SUN will contribute to the proper organization, management and realization of experimental and theoretical activities. In particular, SUN will be focused on bio-refinery aqueous residual cleaning and recycling, by outlining a proper water treatment system.

ENEA: In the framework of VALUEMAG project, collaboration with ENEA will guarantee the presence of strong and valuable scientific background thanks to its experience and core competences concerning energy efficiency, renewable energy sources and innovative technologies. For VALUEMAG project ENEA is committed mainly with three research center, Casaccia Research center that is located in Rome where are working mainly in the characterizations, Trisaia research centre where are located bench and pilot scale for extraction of bioproducts from biomass and at last Portici research centre, in Naples where is a pool of engineering that working on the process development and modelling fields.ENEA is an Associate Member of the Bio-based Industries Consortium (BIC).

NOMASICO: has a strong impact in material processing. NOMASICO has the patent and the technology on electro-spark deposition and therefore on electroporation. Furthermore, NOMASICO has got experience in electromagnetic forming, thus aiding NTUA in developing the soft magnetic cone (SOMAC). The main activitiy of NOMASICO will be the design, development and operation of the pilot plant.

THERACELL: has a history in cosmetics, nutraceuticals and hygienic food. THERACELL is an active SME in developing new pharmaceutical products and is particularly interested and skilled in treating nanoparticles to be insteretd in various types of cells, including stem cells. This way, THERACELL will lead the task in industrialization of the cosmetics and nutraceuticals separation from the micro-algae.

FU SAV: has a vast experience in structural characterization. A number of transmission electron microscopes, with their flagship, Triton, is used to look in details materials in the atomic level, thus being able to perform correlation of structure and properties. Furthermore, the collaborating group of FU SAV has got a large variety of X-ray diffraction machines, DTA facilities and computer cluster, in order to fully characterize materials like those needed in the VALUEMAG project.

PNO: PNO has a strong client base and track record in the interlinked chemical industry, biotech and ecoinnovation sectors, with a particular insight in EU policies and programmes, as well as innovation and funding projects and programmes in 6 European countries. Being partner in multiple EU-funded (including FP7 and Horizon 2020) projects and several commercial assignments, PNO Consultants fully qualifies as a relevant and positive long-term stakeholder within the European BioEconomy panorama. For example, PNO is active partner in the FP7 bio-projects BIO-TIC and BIO-MIMETIC as well as in the Horizon 2020 bio-projects BIOLINX and ROBOX. On a commercial base, PNO has recently produced a practical guide to support the Biobased Industry crossing bridges and optimizing funding possibilities. To help this sector make maximum use of the synergy possibilities, the Bio-based Industries Consortium (BIC) and PNO Consultants (PNO) published the practical guide “Combining BBI (H2020) and European Structural and Investment Funds (ESIF) to deploy the European bioeconomy”. In addition to that, PNO is involved as advisor in roadmap development for the chemical industry towards Public Private Partnership Programmes in HORIZON 2020 (eg: SPIRE), with an important focus on industrial biotech, as well as European networking and incubator projects with various smart specialisation regions which will provide a real life experience base. Finally, PNO has supported the European Commission DG GROWTH Expert Group on BioBased Products on the evaluation of a continental scale Biobased Product Public Procurement Program.

IRIS as an advanced engineering company with a solid expertise in control and monitoring solutions as well as data management and intelligence will undertake the leadership of Subtask 2.1.1 and will

participate in the system integration in task 2.4 under the greenhouse development Work Package. IRIS will also take part of WP4 where will collaborate in modelling tasks for a totally integrated system. IRIS will finally also participate in the Exploitation strategy to develop a successful business route for the Project Results (WP5).

EXERGY: has a strong experience in designing and modelling chemical and biochemical process, selecting the best alternative among different solutions. Their background is in the chemical engineering and environmental process area. In VALUEMAG project, EXERGY will participate in the identification and selection of the best alternatives for CO2 and water remediation from SC gasification and will be part of the engineering design modelling and simulation of the innovative processes. EXERGY is full Member of the Bio-based Industries Consortium (BIC).

VERTECH is a French SME engineering with main expertise on environmental engineering and life cycle thinking. Currently, Vertech participates in 12 European R&D projects (seven FP7 projects and five Horizon 2020 projects). VERTECH will give value to the proposal by its high-tech environmental evaluation, quantification of environmental benefits and selection of best alternatives form the environmental point of view.

EPARELLA is a global technology leader in the construction of algaculture systems. Its worldwide-patented process enables a continuous, industrial-scale production of biomass with the greatest possible conservation of resources and the environment. EPARELLA specialises in the research, development and operation of efficient systems for the cultivation of microorganisms on an industrial scale.

The structure of the consortium demonstrates all the necessary and sufficient characteristics to guarantee a successful project. As visible from the following table the project will harness a variety of competences, all essential to the success of such a project.

Competences	NTUA	SUN	ENEA	NOMASICO	THERACELL	FU SAV	PNO	IRIS	EXERGY	VERTECH	EPARELLA
Developing Magnetic nanoparticles	✓										
Performing electroporation				✓							
Structural characterization						✓					
Magnetic driving and trapping	✓										
Cosmetics, foods, nutraceuticals			✓		✓						
LCA – Modelling		✓							✓	✓	
System automations - Integration								✓			✓
Business models and market strategies				✓			✓	✓			✓
Dissemination and promotion initiatives	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IPR issues	✓	✓	✓			✓					
BIC Members			✓						✓		

Table 3.1: Competence map in VALUEMAG

Figure 3.5, outlines the roles of VALUEMAG partners and their responsibilities. The same figure provides also the connections (supply chain) between the roles and indicates the project beneficiaries selected to fulfill these.

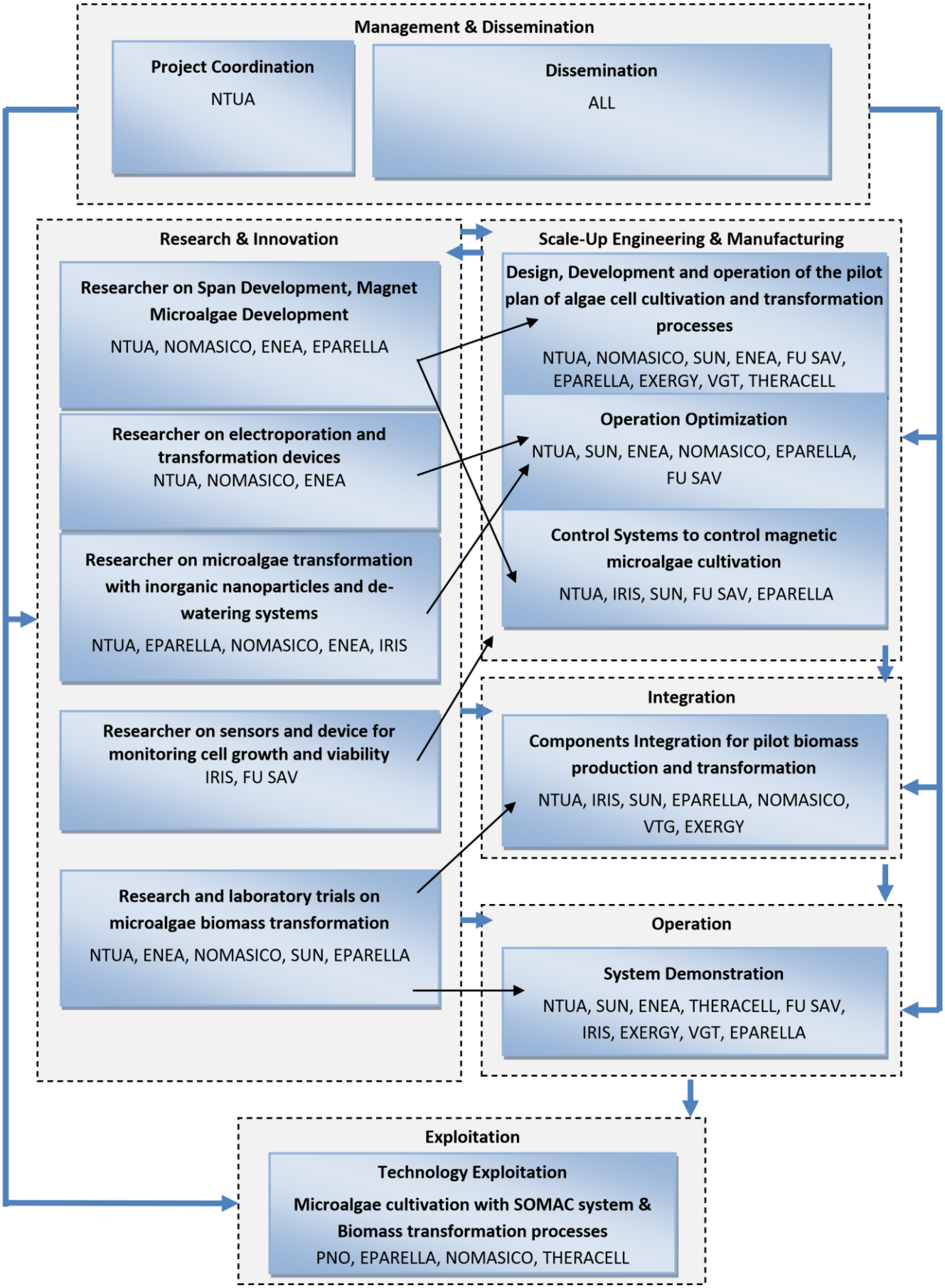

Management & Dissemination
Project Coordination
NTUA
Dissemination
ALL
Research & Innovation
Researcher on Span Development, Magnet Microalgae Development
NTUA, NOMASICO, ENEA, EPARELLA
Researcher on electroporation and transformation devices
NTUA, NOMASICO, ENEA
Researcher on microalgae transformation with inorganic nanoparticles and de-watering systems
NTUA, EPARELLA, NOMASICO, ENEA, IRIS
Researcher on sensors and device for monitoring cell growth and viability
IRIS, FU SAV
Research and laboratory trials on microalgae biomass transformation
NTUA, ENEA, NOMASICO, SUN, EPARELLA
Scale-Up Engineering & Manufacturing
Design, Development and operation of the pilot plan of algae cell cultivation and transformation processes
NTUA, NOMASICO, SUN, ENEA, FU SAV, EPARELLA, EXERGY, VGT, THERACELL
Operation Optimization
NTUA, SUN, ENEA, NOMASICO, EPARELLA, FU SAV
Control Systems to control magnetic microalgae cultivation
NTUA, IRIS, SUN, FU SAV, EPARELLA
Integration
Components Integration for pilot biomass production and transformation
NTUA, IRIS, SUN, EPARELLA, NOMASICO, VTG, EXERGY
Operation
System Demonstration
NTUA, SUN, ENEA, THERACELL, FU SAV, IRIS, EXERGY, VGT, EPARELLA
Exploitation
Technology Exploitation
Microalgae cultivation with SOMAC system & Biomass transformation processes
PNO, EPARELLA, NOMASICO, THERACELL

Figure 3.5: VALUEMAG Supply Chain showing fit of partners to the Project